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Comprehensive evaluation of harmonization on functional brain imaging for multisite data-fusion



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ABSTRACT

To embrace big-data neuroimaging, harmonizing the site effect in resting-state functional magnetic resonance imaging (R-fMRI) data fusion is a fundamental challenge. A comprehensive evaluation of potentially effective harmonization strategies, particularly with specifically collected data, has been scarce, especially for R-fMRI metrics. Here, we comprehensively assess harmonization strategies from multiple perspectives, including tests on residual site effect, individual identification, test-retest reliability, and replicability of group-level statistical results, on widely used R-fMRI metrics across various datasets, including data obtained from participants with repetitive measures at different scanners. For individual identifiability (i.e., whether the same subject could be identified across R-fMRI data scanned across different sites), we found that, while most methods decreased site effects, the Subsampling Maximum-mean-distance based distribution shift correction Algorithm (SMA) and parametric unadjusted CovBat outperformed linear regression models, linear mixed models, ComBat series and invariant conditional variational auto-encoder in clustering accuracy. Test-retest reliability was better for SMA and parametric adjusted CovBat than unadjusted ComBat series and parametric unadjusted CovBat in the number of overlapped voxels. At the same time, SMA was superior to the latter in replicability in terms of the Dice coefficient and the scale of brain areas showing sex differences reproducibly observed across datasets. Furthermore, SMA better detected reproducible sex differences of ALFF under the site-sex confounded situation. Moreover, we designed experiments to identify the best target site features to optimize SMA identifiability, test-retest reliability, and stability. We noted both sample size and distribution of the target site matter and introduced a heuristic formula for selecting the target site. In addition to providing practical guidelines, this work can inform continuing improvements and innovations in harmonizing methodologies for big R-fMRI data.

1. Introduction

In the past decade, brain imaging has entered the era of big data, engendering opportunities and challenges (Charles et al., 2020; Li et al., 2019). Large-scale studies enhance statistical power and decrease false positive rates relative to single-site small-sample studies (Marek et al., 2022), making it possible to detect reliable albeit subtle differences (Button et al., 2013; Varoquaux, 2018).

Brain imaging big data roughly break down into two types: 1) single-site accumulation, represented by the Human Connectome Project (HCP, (Toga et al., 2012)) and the UK Biobank (Sudlow et al., 2015); and 2) multi-site aggregation, which can be divided into prospective cohorts like the Adolescent Brain Cognitive Development Study (ABCD, (Volkow et al., 2018)) and retrospective cohorts such as Enhancing Neuro Imaging Genetics Through Meta-Analysis (ENIGMA, (Thompson et al., 2014)) and the Depression Imaging REsearch ConsorTium (DIRECT, (Chen et al., 2022b; Yan et al., 2019). The former (single-site accumulation) is costly in both time and labor required, while the latter (multi-site aggregation) inevitably entails confounders from multi-site pooling. Even the former approach is likely to encounter hardware and software upgrades, acquisition variations, and other factors, all of which contribute to "site effects" (Yan et al., 2013). Site effects can obscure neuroimaging results of interest, producing false positive results and decreasing statistical power, undermining the credibility and generalizability of purported discoveries (Dansereau et al., 2017; Fortin et al., 2018; Yamashita et al., 2019).

Site harmonization removes or compensates for site effects, mainly through post hoc correction. There are few options for site harmoniza-

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tion of resting-state functional magnetic resonance data (R-fMRI). As a non-invasive technique that allows mapping functional circuits without an explicit task (Biswal et al., 1995), R-fMRI has become a mainstream approach in the study of cognitive neuroscience and psychopathology. In 2013, Yan et al. proposed removing R-fMRI site effects by drawing on methodologies from microarray gene expression. This effort included parametric and nonparametric standardization techniques using either additive or additive and multiplicative approaches for R-fMRI metrics, which were applied to The 1000 Functional Connectomes Project dataset (FCP (Biswal et al., 2010)). This influential work underlined the importance of standardizing site effects and suggested post hoc standardizations of mean-centering and variance standardization. However, as R-fMRI data harmonization techniques have advanced, we undertook an up-to-date comprehensive evaluation to update recommendations on data analysis best practices and to identify needed future innovations for big-data science.

Standard evaluation procedures for a harmonization methodology should address at least two desiderata - removing site-wise heterogeneity while retaining biologically meaningful variability (Fortin et al., 2017). For the former requirement, the non-significance of site statistical tests after harmonization is the generally accepted criterion. Various methods have been proposed for the latter (retaining biological variation). For example, Fortin et al. (2018) suggested that an increased proportion of variation explained by age and improved accuracy in multivariate prediction of age by harmonized measures could reflect the preservation of biological variability, thus validating the effectiveness of the approach. These strategies imply that the brain's biological signal of interest is colinear with the age signal. Any non-colinear signals are not guaranteed to be preserved. Additionally, this approach still lacks the ground truth of brain-wise representations for age. Considering the uncertainty of multi-voxel patterns for any demographic feature, we take identifiability of individuals as an objective indicator validating the conservation of biological variation in the present study. Specifically, we leveraged a neuroimaging dataset containing 41 subjects who traveled across three scanning sites (Chen et al., 2020). We assessed whether clustering accuracy (data from the same subject across different sites clustered together) of harmonized brain-wise R-fMRI metrics would be enhanced, reflecting reliable within-subject resting-state functional organization after filtering out site effects.

One fundamental feature of science is test-retest reliability, which is the ability of an entire experiment to be reproduced, from data acquisition to conclusions. Findings from studies with small statistical power can be hard to replicate (Ioannidis, 2005). Fortunately, low statistical power can be compensated by large sample-size datasets (Marek et al., 2022). Accordingly, how different harmonization methodologies affect the test-retest reliability of multicenter studies has been critical for big data research (Zuo and Xing, 2014; Zuo et al., 2019) but has not been emphasized in previous evaluations. Sex as a primary biological phenotype has been investigated thoroughly using R-fMRI with relatively large-scale samples (Allen et al., 2011; Biswal et al., 2010; Ritchie et al., 2018). Thus, we evaluated the test-retest reliability of sex differences on R-fMRI measures by quantifying overlapped statistical results. Specifically, we used the Dice coefficient to characterize the replicability of sex differences between two scanning sessions in the same population. The sample was compiled from the Consortium for Reliability and Replicability (CORR, Zuo et al., 2014), a multi-site, large-scale dataset. To draw a replicated conclusion, findings from a given study must be translated to different experimental conditions, such as distinct populations, experimental designs, analyses, and research tools. Accordingly, we further evaluated the replicability of sex differences via the overlap of statistically significant results between CoRR and FCP. We hypothesized that larger Dice coefficients and a considerably higher number of significant voxels indicate better data quality harmonization by revealing reliable biological representations.

ComBat is a prevailing method for harmonizing site effects. First introduced in the gene expression microarray field (Johnson et al., 2007), it decomposes site effects into additive and multiplicative components and iteratively estimates their hyperparameters with methods of moments to compute the posterior additive and multiplicative effects for each site. It is notable for its robustness to outliers. Therefore, ComBat can handle aggregation with small sample-size datasets and is conveniently applicable. However, when estimating site effects, it does not take biological effects into account. Instead, they are regarded as fixed, which can be confounded with technical variation partially or wholly (Beer et al., 2020; Chen et al., 2022a; Pomponio et al., 2020). Zindler et al. (2020) reported that sample distribution and usage of covariates affected the false positive rate when ComBat was applied. Besides ComBat, various strategies were proposed to resolve site effects along the data fitting spectrum. Here we examine both parametric methods (i.e., linear regression, linear mixed regression, and parametric ComBat, Correcting Covariance Batch Effects, CovBat) and nonparametric approaches (i.e., nonparametric ComBat, a subsampling maximum-mean-distance based distribution shift correction algorithm, SMA, and a representative of deep learning - invariant conditional variational autoencoder, ICVAE) (see Table 1). Comparing these mainstream harmonization methodologies would yield new insights for big data analyses.

We took five commonly used R-fMRI metrics to evaluate the site effect correction strategies mentioned above. Three of them characterize regional activity features, namely amplitude of low frequency fluctuation (ALFF), the fraction of amplitude of low frequency fluctuation (fALFF), and regional homogeneity (ReHo). One characterizes intrinsic global features, namely degree centrality (DC). The final is the functional connectivity matrix (FC), which denotes brain activity from multiple perspectives. All metrics used in the present study can be easily calculated with the DPARSF pipeline (Yan and Zang, 2010).

Here, we used two test-retest datasets and a large subset of FCP to evaluate the performance of all harmonization methods (Fig. 1): 1) 41 subjects traveling across three scanners (Traveling Subject Project, TSP-3) (Chen et al., 2020); 2) Pooling sites with participants scanned and rescanned within the same scanner, including 420 subjects from CoRR; 3) Six hundred subjects from 15 FCP sites. We utilized TSP-3 to investigate individual-level validity and reliability by applying the identification test. CoRR and FCP were used to evaluate the test-retest reliability and replicability (i.e., external validity) of group-level biological effects by quantifying the Dice coefficient. First, we checked the efficiency of removing voxel/edge-wise site effects through ANOVA for each dataset. We visualized harmonized results by examining whole-brain *P*-value distributions and statistical outputs. We also employed multivariate tests to see if site prediction was less accurate after applying the site effect correction strategy. Second, we assessed the individual identifiability by divisive clustering on these measures with TSP-3 to see if harmonization methods could preserve individual biologically meaningful information. Next, we examined how harmonization approaches affected the test-retest reliability of biological differences on the same sample with test-retest sessions from CoRR. Then, we examined the extent to which biological patterns overlap between CoRR and FCP after harmonization. Finally, we designed an extreme case where site and sex are "perfectly" confounded to examine the coping capacity of methods. Based on tests on residual site effect, identifiability, test-retest reliability, replicability, and dealing with extreme cases, we concluded that SMA outperformed in retaining individual-level and group-level distinctions while eliminating site effects. Accordingly, we undertake experiments to optimize SMA to ensure the best identifiability, test-retest reliability, and stability given the available datasets. We also intended to provide practical advice and an algorithm for target site selection.

2. Methods

2.1. Datasets

The TSP-3 dataset included 41 (22 females; mean age = 22.7 ± 4.1 years; 40 right-handed and one left-handed) young, healthy adults with

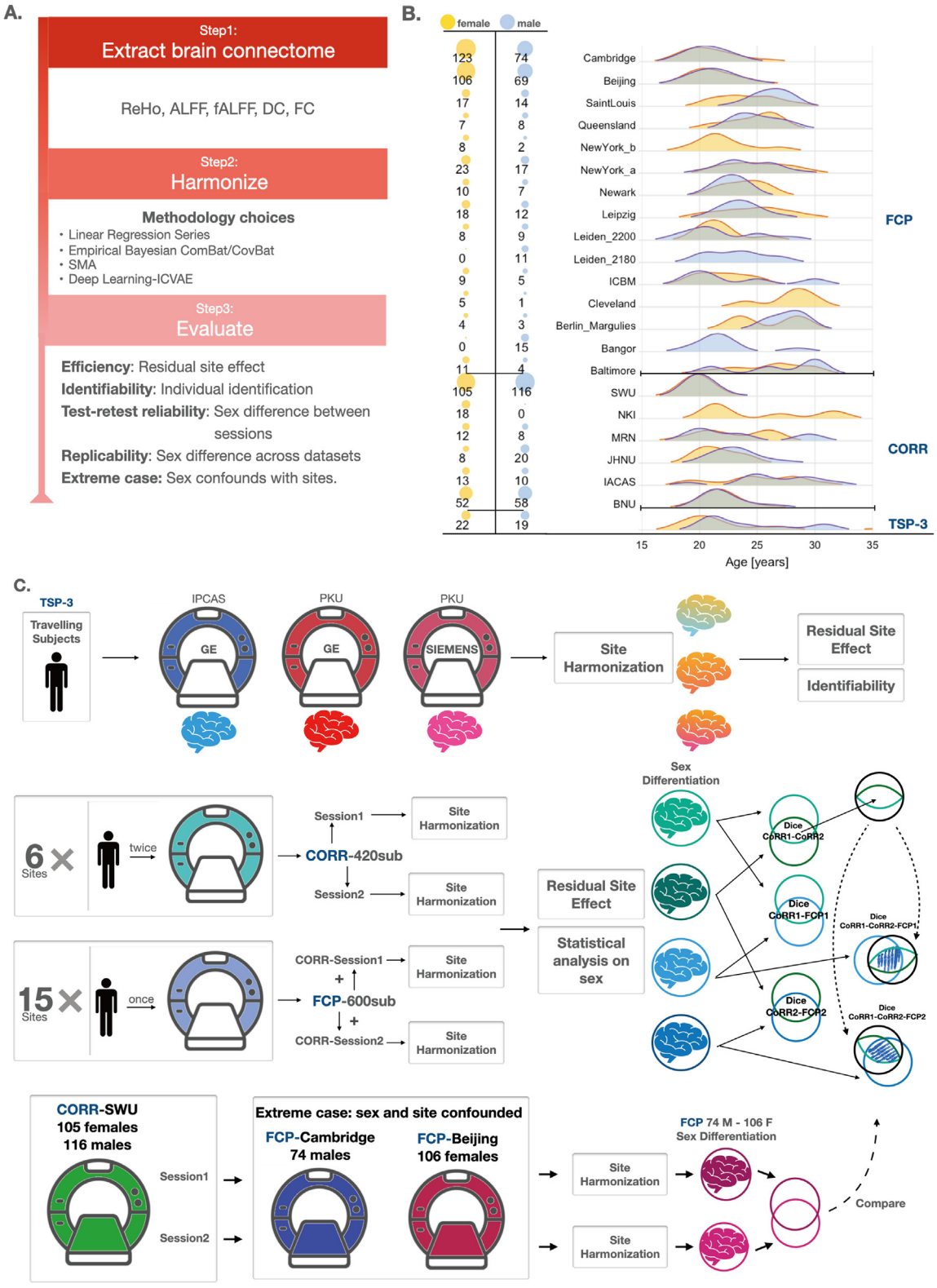


Fig. 1. Illumination of evaluations and demographic information. (A) Evaluation framework. We extracted five widely used R-fMRI metrics of three datasets - TSP-3, CoRR and FCP (Step 1). Then we harmonized their site effects with 10 methods, respectively (Step 2). We assess these methods from five perspectives: residual site effect in mean, variance and covariance, individual identifiability, test-retest reliability, replicability and an extreme case where site and sex are fully confounded (Step 3). (B) Demographic information of the datasets used in this work. This includes dataset and site names, age distributions and numbers of females and males of each site. (C) The evaluation processes. All harmonization of site effects was tested first. Identification was run at an individual level on 41 subjects traveling among three sites. We used divisive clustering to cluster the 123 images and see whether the 3 images belonging to each subject could be correctly grouped together after harmonization (first row). Group-level statistical analysis on sex (Male vs Female): Dice coefficients between repeated scans of CoRR (test-retest reliability, second row) and across CoRR and FCP (replicability, third row), and results under the extreme case where site and sex were fully confounded (fourth row). Also shown are how to interpret each set of Dice coefficients.

Table 1
Methodology spectrum involved in the evaluation. X means yes.

Harmonization Methodology			Addresses additive noise	Addresses multiplicative noise	Addresses Covariance	Addresses nonlinear effects	Non-parametric	Formular
Linear regression	without covariates		x					$Reg: y = \alpha + X_s\beta_s + \varepsilon$
	with biological covariates		x					$Adj: y = \alpha + X_b\beta_b + X_s\beta_s + \varepsilon$
	mixed model with covarites		x					$LMM: y = \alpha + X_b\beta_b + Z_s\mu_s + \varepsilon$
ComBat	parametric	Adjusted	x	x				$y^* = \frac{\sigma_m}{\hat{\delta}^*_{im}}(Z_{ijm} - \hat{\gamma}^*_{im}) + \hat{\alpha} + X_b\hat{\beta}_b$
		Unadjusted	x	x				$y^* = \frac{\sigma_m}{\hat{\delta}^*_{im}}(Z_{ijm} - \hat{\gamma}^*_{im}) + \hat{\alpha}$
	nonparametric	Adjusted	x	x			x	$y^* = \frac{\sigma_m}{\hat{E}(\delta_{im})}(Z_{ijm} - \hat{E}(\gamma_{im})) + \hat{\alpha} + X_b\hat{\beta}_b$
		Unadjusted	x	x			x	$y^* = \frac{\sigma_m}{\hat{E}(\delta_{im})}(Z_{ijm} - \hat{E}(\gamma_{im})) + \hat{\alpha}$
CovBat			x	x	x			$y^* = (y_{ComBat} - e_{ComBat}) + \sum_{k=1}^{K}(\lambda_k\Phi_k)_{ComBat}\Phi_k^T + \sum_{l=K+1}^{\min(N_{sub}, N_{feature})}\lambda_l\Phi_l\Phi_l^T$
Subsampling Maximum-mean-distance based Algorithm			x	x	x	x	x	$y^*_{source} = SMA(argmin\lambda \in \Omega_\lambda \widehat{MMD}(h^\lambda(y_{source}), y_{target}))$
Invariant Conditional Variational Auto-Encoder			x	x	x	x	x	$\overrightarrow{y^*} = ICVAE(\vec{y})$

normal or corrected to normal vision from the community. The Institutional Review Board of the Institute of Psychology, Chinese Academy of Sciences approved experimental procedures. All participants provided informed consent.

Participants traveled amongst three scanners. Images were acquired on two 3 Tesla GE MR750 scanners at the Magnetic Resonance Imaging Research Center, Institute of Psychology, Chinese Academy of Sciences (henceforth IPCAS-GE) and Peking University (henceforth PKU-GE) as well as one 3 Tesla SIEMENS PRISMA scanner (henceforth PKU-SIEMENS) at Peking University. Images were acquired with 8-channel coils for IPCAS-GE and PKU-GE, and a 64-channel coil for PKU-SIEMENS. Before functional image acquisitions, all participants underwent a 3D T1-weighted scan first (IPCAS/PKU-GE: 192 sagittal slices, TR = 6.7 ms, TE = 2.90 ms, slice thickness/gap = 1/0mm, in-plane resolution = 256 × 256, inversion time (IT) = 450ms, FOV = 256 × 256 mm, flip angle = 7º, average = 1; PKU-SIEMENS: 192 sagittal slices, TR = 2530 ms, TE = 2.98 ms, slice thickness/gap = 1/0 mm, in-plane resolution = 256 × 224, inversion time (TI) = 1100 ms, FOV = 256 × 224 mm, flip angle = 7º, average=1). After T1 image acquisition, functional images were scanned (IPCAS/PKU-GE: 33 axial slices, TR = 2000 ms, TE = 30 ms, Flip angle = 90º, Slice thickness/gap = 3.5/0.6 mm, FOV = 220 × 220 mm, Matrix = 64 × 64; PKU-SIEMENS: 62 axial slices, TR = 2000 ms, TE = 30 ms, Flip angle = 90º, Slice thickness/gap = 2/0.3 mm, Slice acceleration = 2, FOV = 224 × 224 mm). A multiband scan (slice acceleration = 2) was performed on PKU-SIEMENS to obtain a voxel size of 2 × 2 × 2 mm. This dataset has been openly shared online (http://rfmri.org/RuminationfMRIData or http://doi.org/10.57760/sciencedb.o00115.00002).

The CoRR dataset originally included 549 subjects scanned twice in the same scanner (mean interval = 205 ± 161 days). From that dataset, 420 subjects (208 females) were selected after quality control with the following exclusion criteria. Only young adults (aged 18 to 30 years old) were included to avoid confounding development. Subjects were excluded if their functional scans showed excessive motion (mean FD_Jenkinson ≥ 0.2 mm) (Jenkinson et al., 2002a). Participants with poor T1 or functional images, low-quality normalization, or inadequate brain coverage were also excluded. Further information on CoRR, including scanning protocols, can be found at http://fcon_1000.projects.nitrc.org/indi/CoRR/html/index.html.

FCP aims at characterizing the reproducibility of R-fMRI across imaging centers. Data of FCP have been openly shared through Neuroimaging Informatics Tools and Resources Clearinghouse (NITRC; http://www.nitrc.org). After applying the same selection criteria for the CoRR dataset, we excluded all subjects scanned in 1.5T scanners. In the end, we used 600 subjects (349 females) from 15 sites. Unlike CoRR, most of the sites chosen from FCP were located in western countries, likely with more western participants. We had to ignore differences in ethnicity and culture in our analyses as only sex, age, and imaging center information are provided (http://fcon_1000.projects.nitrc.org/fcpClassic/FcpTable.html). We calculated age differences among CoRR and FCP sites, and found the significant age differences across sites for both datasets ($P < .001$). The sample sizes and age distributions by sex from each site for all datasets are shown in Fig. 1B.

TSP-3 is a dataset comprised of traveling subjects, which is especially suitable for site prediction. The composition of samples is identical for all three sites. As a result, the most salient factor that affects the validity and reproducibility of findings would be the site effect. CoRR is a test-retest dataset where each subject was scanned twice within one scanner. Site effects and sample differences (sex, age, etc.) vary across sites. The site effect affects groups of subjects, not individuals. As a result, it is not suitable for the identity test. Nevertheless, we could still characterize test-retest reliability using this dataset. Although lacking traveling subjects or repeated measures, FCP has a large total sample size (N = 600). We intended to see whether statistically significant sex differences from CoRR would generalize to such a large sample-sized dataset (Table 2).

2.2. Preprocessing

Unless otherwise stated, we applied the Data Processing Assistant for Resting-State fMRI [(Yan and Zang, 2010), http://R-fMRI.org/DPARSF] for pre-processing, which is based on the Statistical Parametric Mapping (Friston et al., 1994a) (http://www.fil.ion.ucl.ac.uk/spm) and the toolbox for Data Processing & Analysis of Brain Imaging [DPABI (Yan et al., 2016), http://R-fMRI.org/DPABI]. The initial five volumes for TSP-3 and ten for CoRR as well as FCP were discarded to allow data to reach equilibrium, following a slice-timing correction. The realignment on functional volumes used a six-parameter (rigid body) linear transformation with a two-pass procedure (registered to the first image and then registered to the mean of the images after the first realignment). After the realignment, individual T1-weighted MPRAGE images were co-registered to the mean functional image using a 6-degree-of-freedom linear transformation without re-sampling and then segmented into gray matter, white matter (WM), and cerebrospinal fluid (CSF). Finally, transforming images from individual native space to MNI space with the Diffeomorphic Anatomical Registration Through Exponentiated Lie algebra (DARTEL) tool (Ashburner, 2007). We regressed WM and CSF signals to reduce respiratory and cardiac effects, and choose the Friston 24-parameter model (Friston et al., 1996) (i.e., six head motion parameters, six head motion parameters one time point before, and the 12 corresponding squared items) to minimize head motion confounds. This model was chosen because higher-order models can control head motion

Table 2
Illustration of the relationship and roles in evaluation for different datasets. "x"-marked dataset (row) independently participated in that evaluation item (column); "+"-marked datasets were combined in that evaluation. The first 4 evaluations were for all methods. "Site and sex confounded" is in italics for it was only applied on ComBat, CovBat, and SMA of ALFF; "Replicability" is in italics for it was only applied on raw, adjusted ComBat, parametric-adjusted CovBat, and SMA; "Target Site choice (SMA)" is in italics for it was only applied on SMA, while all other evaluations were performed on all methods.

Role in evaluation	Residual site effect	Site prediction	Identifiability	Test-retest reliability	*Site and sex confounded*	*Replicability*	*Target Site choice (SMA)*
TSP-3	x	x	x				x
CoRR	x			x	+	+	x
FCP	x				+	+	x

effects better (Satterthwaite et al., 2013; Yan et al., 2013). Additionally, linear trends were included as a regressor to account for drifts in the blood oxygen level dependent (BOLD) signal. Due to the continuing controversy, we did not perform global signal regression (Murphy and Fox, 2016). Temporal bandpass filtering (0.01-0.1Hz) was completed except for ALFF and fALFF. Of note, temporal bandpass filtering was performed after nuisance regression so as not to reintroduce nuisance-related variation (Hallquist et al., 2013).

2.3. R-fMRI Metrics

ALFF (Zang et al., 2007) and fALFF (Zou et al., 2008): ALFF is the mean of amplitudes within a specific frequency domain (here, 0.01–0.1Hz) from a fast Fourier transform of a voxel's time course. fALFF is a normalized version of ALFF and represents the relative contribution of specific oscillations to the whole detectable frequency range.

ReHo (Zang et al., 2004): ReHo is a rank-based Kendall's coefficient of concordance that assesses the synchronization among a given voxel and its nearest neighbors' (here, 26 voxels) time courses.

DC (Buckner et al., 2009; Zuo et al., 2012): DC is the number or sum of weights of significant connections for a voxel. Here, we calculated the weighted sum of positive correlations by requiring each connection's correlation coefficient to exceed a threshold of $r > 0.25$ (Buckner et al., 2009).

To define the area for voxel-wise analyses, we first created a mask representing the overlap area for TSP-3, CoRR, and FCP with a 90% group mask (i.e., the voxels selected for the group mask are included in the EPI automasks of 90% of the subjects), containing 38810 voxels. We then extracted each metric based on the unified mask for harmonization.

FC is defined as the temporal coincidence of spatially distant neurophysiological events (Friston, 2011). We selected a total of 142 cortical regions of interests (ROIs) from the Dosenbach 160 atlas (Dosenbach et al., 2010). Each ROI corresponded to 7 large-scale brain networks: the default mode network (DMN), sensory-motor network (SMN), dorsal attention network (DAN), ventral attention network (VAN), subcortical network (SC, instead of the limbic network), visual network (VN) and frontal-parietal network (FPN). ROIs in the cerebellum were excluded. We then computed Pearson's coefficients between each paired ROIs' mean time-series.

Before entering into further analyses, all voxel-wise metric maps were Z-standardized (subtracting the mean value for the entire brain from each voxel and then dividing by the whole-brain standard deviation), and functional images were smoothed with a 4mm Gaussian kernel. A Fisher's R to Z transformation was applied to the FC matrix.

2.3. Harmonization Methods

2.3.1. Linear regression

The linear regression based methods assumed the site factor to be an additive effect on data. This effect is normally distributed and orthogonal to other potential factors. Of note, this assumption might not be true, especially in the FCP dataset, because demographic differences (like ethnicity) were impossible to model. We considered three methods - a linear regression model with (*Adj*) or without biological factors adjusted (*Reg*) to remove additive site effect estimated by ordinary least squares (OLS) and a linear mixed model (*LMM)* with covariates adjusted of which site was treated as a random effect and estimated with restricted maximum likelihood (REML).

$$Reg: y = \alpha + X_s\beta_s + \varepsilon \quad (1)$$

$$Adj: y = \alpha + X_b\beta_b + X_s\beta_s + \varepsilon \quad (2)$$

$$LMM: y = \alpha + X_b\beta_b + Z_s\mu_s + \varepsilon \quad (3)$$

y denotes a voxel/edge-wise R-fMRI metric, and α is the weighted mean of each feature for all sites. Residual terms ε have a mean 0. The critical steps for the above methods include: 1) estimating the coefficients for sites (β_s or μ_s), and 2) subtracting the estimated site effect ($X_s\beta_s$ or $Z_s\mu_s$) from y.

2.3.2. ComBat

The ComBat (combating batch effects when combining Batches) method was originally introduced from the technology of removing batch effects in genetics (Johnson et al., 2007). ComBat has been widely utilized to harmonize MRI data, including DTI (Fortin et al., 2017), cortical thickness (Fortin et al., 2018), and R-fMRI FCs (Yu et al., 2018). It is currently recognized to be a convenient and fast harmonization method. Compared with linear regression, ComBat further deconstructs site effects into additive (location) and multiplicative (scale) components and then adopts the empirical Bayesian to estimate the posterior distribution to compute the mean and variance of site effects.

$$ComBat: y_{ijm} = \alpha + X_b\beta_b + \gamma_{im} + \delta_{im}\varepsilon_{ijm} \quad (4)$$

wherein i stands for the site, j stands for the individual sample index, and m stands for the unit of each metric (voxel or edge here). γ_{im} and δ_{im} represent the estimated additive and multiplicative site effects of the selected feature m (voxel or edge here) of site i, ε_{ijm} is a normally distributed error term with zero mean and variance σ_m.

$$Z_{ijm} = y_{ijm} - \frac{\left(\hat{\alpha} + X_b\hat{\beta}_b\right)}{\hat{\varepsilon}_{ijm}}$$

$$Z_{ijm} \sim N\left(\gamma_{im}, \delta_{im}^2\right)$$

To estimate posterior distributions of standardized site effects for known samples, we deployed both the commonly used parametric method and a nonparametric approach that theoretically matches R-fMRI metrics, for which high dimensional data would be non-normally distributed.

1) For parametric ComBat, the mean and variance of a site effect hypothetically follow gaussian and inverse-gamma distributions.

$$\gamma_{im} \sim N\left(\gamma_i, \tau_i^2\right) \text{ and } \delta_{im}^2 \sim InverseGamma\left(\lambda_i, \theta_i\right)$$

We applied the Methods of Moments to iteratively estimate means and variances until convergence.

2) For nonparametric ComBat, the prior distribution's site-effect probability density function is undefined. Thus, we estimate the posterior expectations of the site effect parameters, denoted $E[\gamma_{im}]$ and $E[\delta_{im}^2]$, respectively, through Monte Carlo Integration.

After obtaining the site effect parameters, the empirical Bayesian was applied to estimate the conditional posterior of the site effect. And finally,

$$y^* = \frac{\sigma_m}{\hat{\delta}^*_{im}}\left(Z_{ijm} - \hat{\gamma}^*_{im}\right) + \hat{\alpha} + X_b\hat{\beta}_b$$

$\hat{\gamma}^*_{im}$ and $\hat{\delta}^*_{im}$ are conditional posterior site effects for each site, and the original Z value of site effect for each individual is multiplied by a normally distributed error term.

2.3.3. Correcting Covariance Batch Effects (CovBat)

Chen et al. (2022a) modified ComBat into CovBat to accommodate machine learning (ML) scenarios. It is intended to eliminate the site effect on the covariance leveraging ML to mine harmonized data. Specifically, it hypotheses within-site covariance has location and scale shifts. Thus, it takes the ComBat-harmonized residuals and transforms them into a covariance matrix by multiplying with the Bayes-estimated variance. Then CovBat decomposes the covariance matrix with principal components analysis (PCA) to obtain the principal components (PCs) and harmonizes them by applying ComBat without covariates, with a tuning parameter K which defines the first K PCs that need within-site harmonization. Finally, they project the adjusted PCs back to the residual space to replace the original ComBat-harmonized residuals.

2.3.4. Subsampling Maximum-mean-distance Algorithm (SMA)

SMA (Zhou et al., 2018) was developed to deal with statistical issues raised by data pooling. SMA is based on a causal graphical model and maximum mean distance (Bareinboim and Pearl, 2016). The idea originates from domain adaptation (Zhuang et al., 2021), a subfield of machine learning. It aimes at shifting the distribution of unseen features/tasks (test set) to the known features/tasks (training set) so that the trained model can handle the unseen features/tasks (the "adaptation" ability). According to this idea, we can break down the general site effect into three factors: population characteristic differences, sample selection bias, and experimental variations (Bareinboim and Pearl, 2016). For conditions where we regard experimental variation as the only factor influencing the distribution of features (e.g., TSP-3 dataset), we can shift the distributions to make them homogeneous. When population differences or sample selection bias occur in different sites, we included these factors in the model. These factors also contributed to the shift of the feature distribution.

SMA needs a specific site to be selected as the standard (here, the "target site") to allow other sites to imitate its site effect. As a result, data from all sites share the same "site tag". SMA constitutes an integrated process from diagnosis to targeted treatment of data. Specifically, if only site-specific nuisance factors are involved, we perform univariate direct shift correction.

$$SMA \ : \underset{\lambda \in \Omega_\lambda}{argmin}\ \widehat{MMD}\left(h^\lambda\left(y_{source}\right), y_{target}\right) \tag{5}$$

where h is an affine function in which λ represents the slope and intercept choice for model selection, which are optimized towards minimization of distance between a univariate feature's distribution between target site and source site; MMD refers to the Maximum Mean Discrepancy, a measure of the difference between two distributions in reproducing Hilbert kernel space (here "distribution" means the probability distributions of features in math). It was first proposed to test whether two probability distributions differ statistically. It defines distance as the most significant difference between two distributions' moments. To obtain the high-order moment, we can use kernel embedding of distributions, which maps the distribution into a point in the reproducing kernel Hilbert space (RKHS). Then we can get the MMD value through the inner product of these two points.

Z variates are those (e.g., sex and age in our hypothesis) that causally impact our measurements (e.g., R-fMRI metrics here) while being variously distributed across sites. We can fit such co-occurring variations (Z covaring with site factor) through a subsampling framework. Explicitly, subgroups are first defined by Z variates with the sample size of each subgroup determined by the distribution of Z variates. When the final model slope and intercepts are selected to minimize MMD between the source site and target site for each voxel/edge, SMA resamples from the corresponding subgroups from both the target site and source site so that identical Z distribution subsamples are constructed for the target and source sites. Under this condition, the only factor that differentiates two distributions is site. Hence we need to find the λ to make distributions of subsamples have the minimum MMD value. We iteratively calculate λ and apply the mean of all models on the entire sample of the source site for each feature.

We used PKU-GE as the target site for the TSP-3 dataset. The scanner's manufacturer of PKU-GE is the same as IPCAS (a GE scanner), while the location is the same as PKU-SIEMENS (Peking University). We chose CoRR-BNU as the target site for the CoRR and FCP datasets as it has a large sample size and it comprises a wide age range for both sexes.

2.3.5. Invariant Conditional Variational Auto-Encoder (ICVAE)

The idea of ICVAE is first to encode site information and the other information separately and use the encoder to learn the distribution characteristics of the original image and abstract them into a low-dimensional Z vector in the intermediate hidden space. Then in the low dimensional space, the site-orthogonal information can be characterized by highly compressed orthogonal dimensions. After decoding, they can be mapped to the same dimensional space with the raw data with the same site label. The aim is to minimize the distance (defined by Kullback-Leibler divergence, KL) (Csiszar, 1975) between the generated image and the original image while minimizing the amount of mutual information between the intermediate representation z and the site information s while controlling the encoding and decoding process to have a certain degree of correspondence. Model implementation details are shown in Fig. S1.

Moyer et al. (2020) were the first to apply ICVAE to harmonize DTI datasets. From deep-learning approaches developed based on domain adaptation (Dinsdale et al., 2021; Zhong et al., 2020), we selected ICVAE for its interpretability.

2.4. Evaluation of Harmonization Effects

2.4.1. Tests on residual site effects

To examine efficiency in removing site-wise heterogeneity, we performed an ANOVA on site differences of R-fMRI metrics after site effect harmonization. Head motion (mean FD Jenkinson, (Jenkinson et al., 2002b)), sex and age were included as covariates. The two sessions of CoRR, the corresponding harmonized FCP (termed FCP1 and FCP2, where numbers mark the CoRR session being harmonized, with 587 subjects selected to match the SMA-harmonized sample size) were all computed separately. We checked if significant site effects of the raw data disappeareded after each harmonization method to evaluate if the removal of mean and variance of site effect worked.

To control multiple comparisons, we performed the Gaussian Random Field (GRF) correction (Friston et al., 1994b) on F statistics across the whole brain at the threshold of one-tailed cluster $P < 0.05$, voxel $P < 0.001$. For functional connectivity matrices, we performed ANOVA at each edge. Multiple comparison correction was corrected by False Discovery Rate (FDR, Genovese et al., 2002) correction at $q < 0.05$. Brain-wise visualization was rendered with the BrainNet Viewer (Xia et al., 2013).

In addition, we applied the random forest method (Matlab2020, TreeBagger, Ntrees = 600) on TSP-3 to investigate the ability to remove covariance (Chen et al., 2022a). Specifically, we adopted leave-one-out cross validation with all features (38810 features for voxel-wise metrics and 10011 features for FC) to predict site. This yielded 123 trained models. We calculated the accuracy of training sets and testing sets separately. Accuracy was defined as the percentage of correct predicted

labels. We compared the 123 training accuracies with Tukey HSD as a post-hoc test (0.05) to control the false positive rate.

2.4.2. Individual identifiability

To test if the harmonization method could retain biologically meaningful individual variability, we used subject identification with divisive analysis (DIANA) (Hadi, 1991) as the clustering methodology. We conducted Euclidean and Pearson algorithms to construct distance matrices for each raw/harmonized metric. We chose five linkage functions, Ward's method (Murtagh and Legendre, 2014) (ward.D and ward.D2 in R), Average, Single and Complete as the agglomeration methods. By first fixing the number of final clusters equal to the number of traveling subjects (N = 41) and blindly clustering individual multivariate intrinsic brain representations, we obtained a 41×41 matrix in which the columns are cluster ID and rows are subject ID. For the TSP-3 cohort, 3 in each matrix element means that the three scans were sorted into the same cluster accurately. Ideally after harmonization, the matrix diagonal would contain only 3's.

We evaluate the performance by counting how many subjects' images would be grouped into a single cluster (i.e., how many 3s in the 41×41 matrix). The accuracy formula is as follows:

$$accuracy = \frac{n}{N_{subjects}}$$

where n denotes the number of subjects whose 3 repeated scans were all grouped into the same cluster and $N_{subjects}$ stands for the number of subjects.

2.4.3. Test-retest reliability: Dice coefficient of brain sex effect between two CoRR sessions

The test-retest reliability reflects whether a measure yields identical results with the same sample, analytical steps, and experimental settings. Here we applied harmonization steps separately for each session of CoRR, and then executed statistical analyses on sex within each session. We took GRF (voxel $P < 0.001$, cluster $P < 0.05$, two tailed) and FDR ($q < 0.05$) as multiple correction methods for voxel-wise brain measures, and FDR ($q < 0.05$) for edge-wise functional connectivity matrices.

Here we used the Dice coefficient (Rombouts, 1998) to quantify between-session replicability. We expected a successful harmonization to improve between-session replicability, reducing confounding site effects.

$$Dice_{CoRR1-CoRR2} = \frac{V_{overlap}}{(V_1 + V_2) \times 0.5}$$

where V_1 means the number of statistically significant voxels for Session1, V_2 means the number of statistically significant voxels for Session2. $V_{overlap}$ symbolizes the number of significant voxels spatially overlapping in both Session1 and Session2.

We applied a non-parametric one-way repeated measures ANOVA (Friedman's test) on the test-retest reliability of five metrics under two correction strategies (FDR and GRF). Post-hoc corrections were performed with Tukey's honest significant difference criterion ($P < .05$).

2.4.4. Replicability: Dice coefficient of CoRR and FCP

Replicability is obtaining consistent results across studies aimed at answering the same scientific question. Here the rationale for harmonizing FCP with CoRR was to investigate whether harmonization during multi-site pooling would enhance the replicability of the original data. We tested replicability by calculating the Dice coefficient across different datasets after harmonization. For SMA, we chose the CoRR BNU site as the target site and harmonized the FCP cohort to it. Age and sex likely affect the organization of spontaneous brain activity (Dosenbach et al., 2010; van Velzen et al., 2020; Zuo et al., 2017). We took 21 years of age (> 21 and ≤ 21) together with binary sex as the bases for subgrouping. For the other methodologies, we combined two datasets (one session of CoRR and FCP) and took age and sex as biological covariates to regress site effects for those adjusted. We then performed statistical analysis on separated harmonized FCP data and executed multiple comparison corrections in the same way as for test-retest reliability analysis.

$$Dice_{between-datasets} = \frac{V_{overlap}}{(V_{D1} + V_{D2}) \times 0.5}$$

Replicability was calculated between 2 datasets, including 1) between CoRR session 1 (CoRR1) and harmonized FCP (hereafter FCP1), i.e., DiceCoRR1-FCP1), 2) between CoRR session 2 (CoRR2) and harmonized FCP (hereafter FCP2), i.e., DiceCoRR2-FCP2, and 3) between the intersection of CORR1∩CoRR2, and FCP1 (DiceCoRR1-CoRR2-FCP1) or FCP2 (DiceCoRR1-CoRR2-FCP2).

Similar to test-retest reliability, we applied a non-parametric one-way repeated measures ANOVA (Friedman's test) on five metrics by two correction strategies (FDR and GRF) and post-hoc analyses corrected by Tukey's honest significant difference criterion were conducted to evaluate chosen methods with regard to replicability.

2.4.5. Extreme case: Perfect confounding between site and sex

To investigate how the methods deal with extreme conditions, we created a toy model where site and group differences were "perfectly" confounded. We chose 106 females from the Beijing site and 74 males from the Cambridge site and took CoRR-SWU (105 female, 116 male) for harmonization. We tested all ComBat series, CovBat, and SMA on ALFF for its superior test-retest reliability across metrics. We calculated the sex difference on the original SWU and harmonized FCP data separately, corrected by GRF (voxel $P < 0.001$, cluster $P < 0.05$, two tailed).

3. Results

3.1. Tests on residual site effects

For unharmonized raw data, a wide range of areas showed significant site differences after GRF correction for all datasets (Fig. 2). Differently distributed site effects were revealed for the same metric across datasets, possibly due to the heterogeneity of scanning settings. For harmonized data, most strategies achieved the goal of removing site effects, i.e., no voxels/edges showed significant site effects after multiple comparison correction and site prediction accuracy was decreased in contrast to that of the unharmonized data.

For TSP-3 (Fig. 2), site effects consistently appeared in most areas of the cortex except the lateral occipital lobe in the unharmonized raw data. The largest F values converged in the posterior cingulate cortex (PCC), lingual gyrus and subcortical areas such as thalamus, caudate and hippocampus. After harmonization, only the ICVAE was unable to completely remove statistically significant areas. For FC, only raw data showed significant edges after FDR correction, primarily within VN, SMN, VAN, and DMN. All methods successfully removed site effects for cortical ROIs.

Significant areas in the unharmonized CoRR data were mainly found in the cingulate, frontal lobe and subcortex for both sessions (Fig. 2). From a voxel-wise perspective, site effects across the two sessions were similarly distributed. The largest F values converged in the frontal lobe and lingual area. After harmonization, ICVAE removed site effects for ReHo and DC and little remained for ALFF and fALFF. For FC, significant site effects were mostly found within and between VN/SMN with all networks in both CoRR sessions. SMA left several edges with significant site effects in CoRR session 2.

We then performed statistical analysis on FCP1/2 to assess whether site effects had been removed. To test replicability, we only harmonized FCP with the best performing methods based on tests of residual site effects, identifiability and test-retest reliability, namely parametric adjusted ComBat (PACB), nonparametric adjusted ComBat (NACB), parametric adjusted CovBat and SMA. From a voxel-wise perspective, unharmonized FCP site effects spanned almost the whole brain except the occipital lobes. The largest F values converged in the temporal lobes and frontal lobes. ReHo and ALFF showed significant differences among

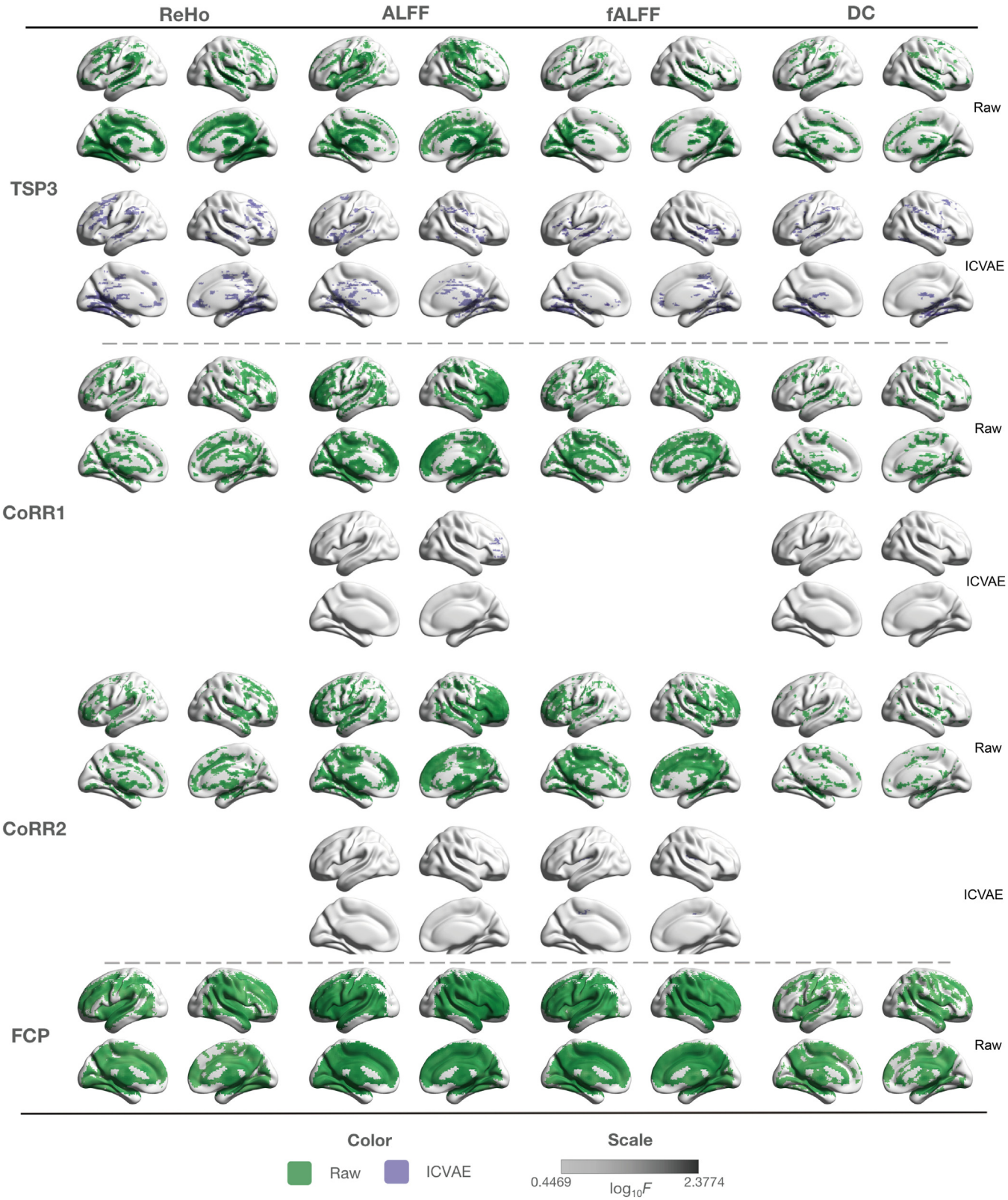


Fig. 2. Residual site effect evaluation - cortical distribution of statistical site effects. We assessed residual site effects with a general linear regression model with age, sex and head motion as covariates for TSP-3, CoRR (session 1 and session 2), and FCP datasets, respectively and illustrate site effects for five R-fMRI metrics. FCP1\2 represents harmonization with CoRR1\2 for FCP. Different colors correspond to different methods. Colored brain area indicates significant results after GRF correction (voxel $P < 0.001$, cluster $P < 0.05$), values equaling $\log_{10}(F)$. Only Raw (first row of each dataset) and ICVAE (the second row of each dataset) results are displayed since only these two retained site effects on the brain surface. The FC heatmap shows the number of edges among large-scale networks with significant site effects after FDR correction ($q < 0.05$). The lower triangle of the heatmap corresponds to session1 results and the upper triangle to session2. Harmonization methods with no voxels/edges with significant site effects after multiple comparison correction are not shown (i.e., only SMA is shown). With the FCP dataset, only adjusted-ComBat/CovBat and SMA are illustrated, as unadjusted-ComBat/CovBat had significantly fewer overlapped voxels and other methods had low identifiability. We note that ICVAE was especially ineffective in harmonizing TSP-3.

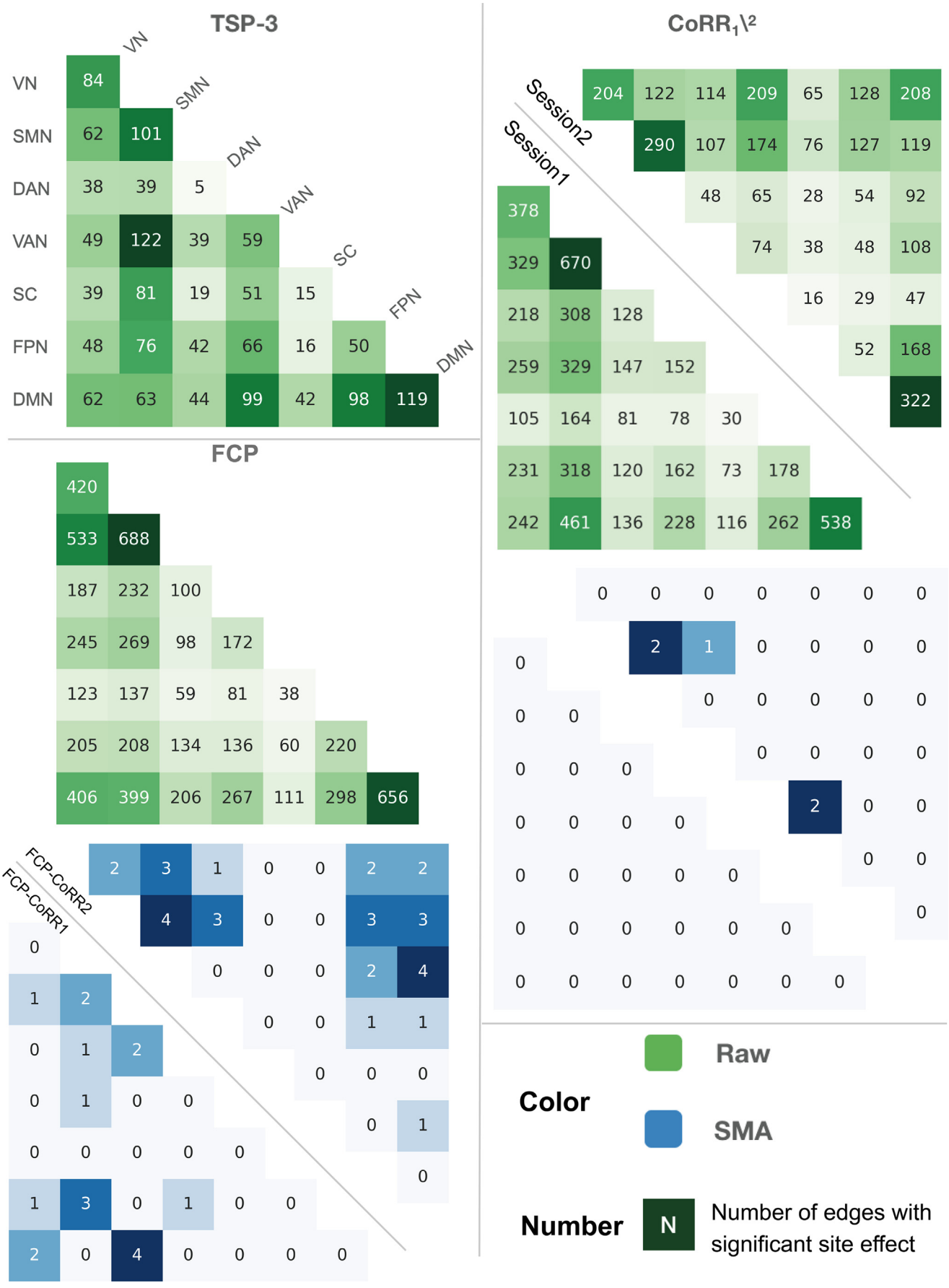


Fig. 2. Continued

Table 3
Permutation test on identifiability. Bold non-italic numbers mean SMA was ranked significantly higher (higher accuracy), and bold italic numbers indicate the reverse (ranked lower).

$Pvalue_{(SMA,\ ComBat\text{-}Series)}$	ComBat (adj+para)	ComBat (adj+nonpara)	ComBat (unadj+para)	ComBat (unadj+nonpara)	CovBat (adj+para)	CovBat (unadj+para)
ReHo	0.03	0.06	0.11	0.05	0.87	***0.01***
ALFF	**0.00**	**0.01**	0.25	0.86	**0.01**	0.03
fALFF	0.02	**0.01**	**0.00**	**0.00**	0.26	0.79
DC	**0.01**	**0.00**	**0.00**	**0.00**	0.20	0.76
FC	0.03	0.03	0.02	0.02	0.04	0.89

different sites in the cerebellum even after site harmonization. No significant differences were found on the cortical surface for all metrics, yet there left significant site effect of ReHo and fALFF in subcortex for SMA. For FC, significant site effects were mostly seen within VN, SMN and DMN, as well as between of their any pairs. Some edges remained significant following SMA in both FCP1/2.

Additionally, we used the random forest algorithm and leave-one-out cross validation to train site classification. SMA's precision was the lowest across all metrics except for ALFF (Fig. 3). Except for ICVAE, the predicted site of the testing image consistently pointed to one of the other two sites instead of the true one, especially for ComBat, CovBat series and SMA, i.e., the accuracies in the left-out testing dataset could reach zero and be far below random chance (33%). This is because we used subject who traveled across 3 sites. If the site effects were present, the test image would be assigned to the correct site (high testing accuracy). However, if the site effects were completely removed, the test image would be assigned to the same subject (the model has seen this subject from the other two sites), thus the site label will always be predicted incorrect (very low testing accuracy). Thus, the closer the accuracy to 0, the better the method completely removed site effects. Post hoc pair-wise comparisons showed that SMA's training accuracy in predicting the site was also significantly lower than other methods except for ALFF (Fig. S3).

3.2. Identifiability

For this evaluation, we wanted to ensure that the appropriate method would remove the right amount of site effect, leaving behind enough effect size to identify the same subject across multiple sites. Hence, we adopted an unsupervised machine learning technique — hierarchical clustering, to divisively cluster all 41 participants' three scans (123 images in total).

We used two kinds of methods (Euclidean and Pearson) to construct distance matrices, and five linkage means (ward.D2, ward.D, average, complete, single) as agglomeration methods (Fig. 4 and Table S6). The random chance of accuracy is around $\frac{\frac{41}{41}\times 1+\frac{40}{41}\times(S^{1}_{123-40\times 3}-1)+\frac{39}{41}\times(S^{2}_{123-39\times 3}-S^{1}_{123-40\times 3})+\cdots}{S^{41}_{123}}\approx 0$ (S denotes Stirling number, we have uploaded this computation code to Github).The accuracy of the linear regression series was better than for raw data except for unadjusted-regressed FC. Their accuracy in ReHo, ALFF, fALFF and DC was lower than ComBat/CovBat and SMA. ICVAE's performance was the worst among all harmonization strategies. In fact, its performance was even worse than the raw data in fALFF, DC, and most in FC.

In order to determine if there is distinction among other methods, we applied permutation tests on ranks for 5 R-fMRI metrics with 10 data points (Euclidean and Pearson 2 distances x 5 linkage construction methods, including ward.D2, ward.D, average, complete, single) to compare each methods with raw and between each pairs of SMA and each ComBat/CovBat model. First, we ranked accuracies of each distance-linkage combination for all metrics (bigger rank means higher accuracy). Next, we took 50000 permutations for each paired comparison to construct the null distribution and performed one-way tests on the rank differences (R 4.2.2 - Coin package- Approximative Two-Sample Fisher-Pitman Permutation Test, one-sided, $P < 0.01$). The results are shown in Table 3, with bold non-italic numbers indicating SMA was ranked significantly higher (larger accuracy), and bold italic number the reverse (ranked lower). As seen in the table, SMA had significantly better identifiability than adjusted ComBat-series in ALFF, fALFF and DC, parametric unadjusted ComBat in fALFF, DC and FC, nonparametric unadjusted ComBat in fALFF, and parametric adjusted CovBat in ALFF; SMA had worse identifiability than parametric unadjusted CovBat in ReHo and ALFF.

In summary, parametric unadjusted CovBat and SMA were generally as accurate or even better than all other methods across metrics. The ComBat series' performance varied across different metrics and strategies (parametric/adjusted). ReHo, ALFF, and FC were more individual-identifiable than fALFF and DC.

3.3. Statistical test-retest reliability

Adjusted linear regression, adjusted ComBat/CovBat and SMA yielded similar DiceCoRR1-CoRR2 values. When examining the number of overlapping voxels, SMA and parametric adjusted CovBat were superior to the others in most cases. Adjusted ComBat typically yielded more overlapped voxels than unadjusted ComBat. These tendencies are revealed in both GRF (voxel $P < 0.001$, cluster $P < 0.05$, Table 4) and FDR ($q < 0.05$, Table S1) corrected results. We found that covariate-adjusted methods, including adjusted linear regression, and adjusted ComBat/Covbat, were much better than unadjusted methods in the number of overlapped significant voxels. Overall, SMA was close in both Dice values and overlapped amounts with covariates-adjusted methods for all metrics. ICVAE had the highest Dice values for CoRR1-CoRR2 and overlapping number of voxels for ALFF, but scored poorly in ReHo and DC. Considering that ICVAE and linear regression series are less appropriate for retaining biological variation based on identifiability, and that unadjusted ComBat/CovBat had less area overlapping across sessions, we focused on adjusted ComBat/CovBat and SMA for the replicability evaluation.

We then applied the Friedman Test on the test–retest reliabilities of between-subject sex on five metrics by two multiple correction strategies (GRF and FDR). The multiple comparisons of the post hoc comparisons were corrected using Tukey's honest significant difference criterion ($P < 0.05$). No significant difference was found in Dice coefficients among all site harmonization methods. However, parametric-adjusted CovBat and SMA retained significantly more overlapped voxels than Reg, LMM, and unadjusted ComBat/CovBat (Fig. 5).

3.4. Statistical replicability

We separately harmonized FCP datasets with each session of CoRR by combining the two datasets (assigning BNU as the target site for FCP in SMA). The results are presented in Table 5, visualized in Fig. 6A (FDR $q < 0.05$) and Fig. S4 (GRF voxel $P < 0.001$, cluster $P < 0.05$). On replicability, PACB and NACB performed similarly. SMA performed better than adjusted ComBat/CovBat series in higher Dice value as well as yielding more replicable results. Of note, 587 subjects underwent SMA analyses, as we excluded Berlin_Margulies (all subjects were over age 23) and Cleveland (all subjects were over age 24 and only 1 was male) as their distributions deviated excessively from the target site [BNU, only 20% subjects were over 23 years old; there should be at least 50% overlap for

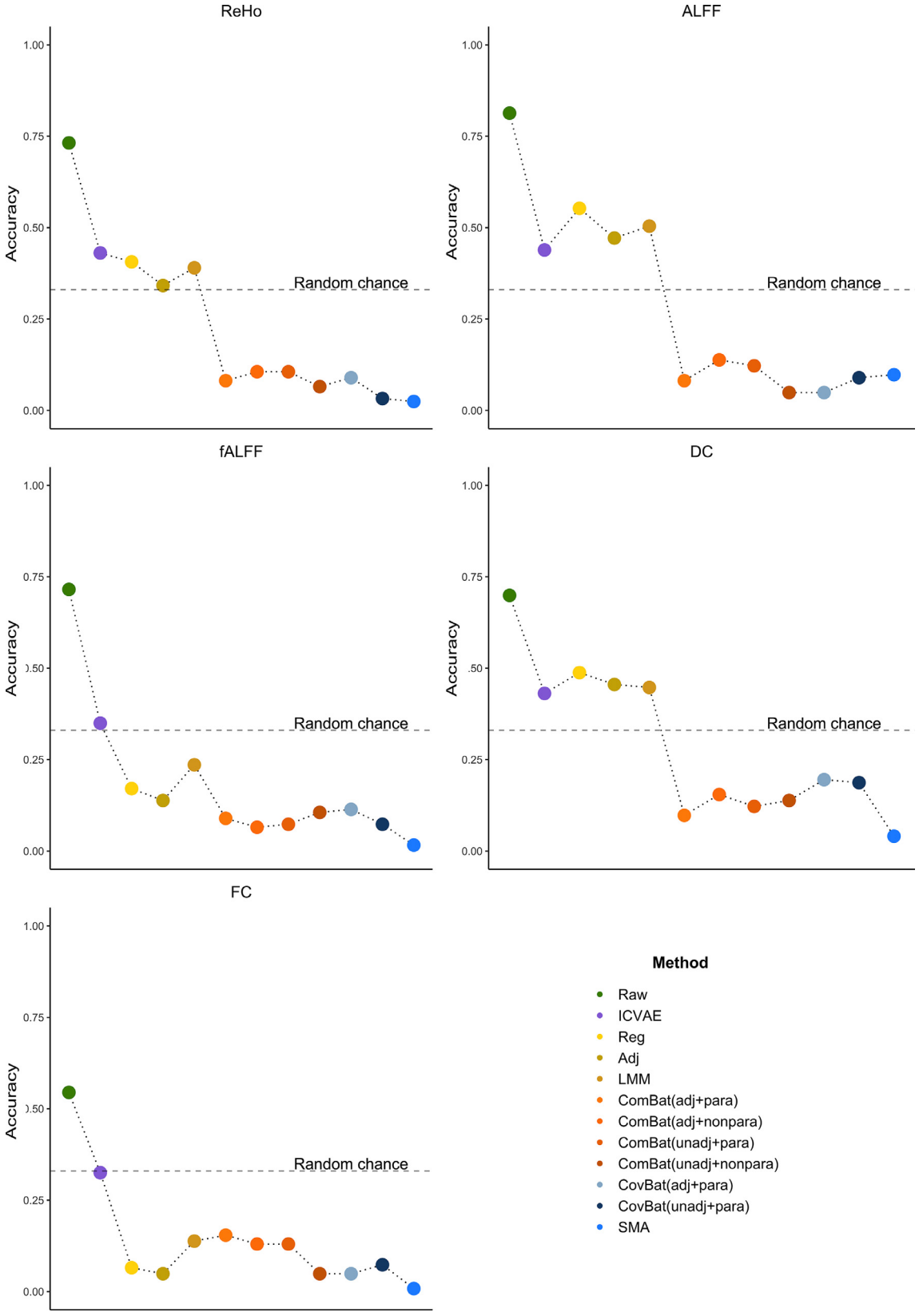


Fig. 3. Residual site effect in covariance – Leave-one-out test accuracy of site classification on TSP-3. Leave-one-out test accuracy of site classification for TSP-3. The points show the precision of testing accuracy of 123 models trained with 122 images of all data and tested on the one image left out for each method. The y-axis is the precision of testing set and the colors distinguish methods. The accuracies in the left-out testing dataset could reach zero and be far below random chance (33%). When the site effects are removed, the test image would be assigned to the same subject the model had already seen from the other two sites, leading to incorrect predictions of site labels and very low testing accuracy. Therefore, the closer accuracy is to zero, the better the method has successfully removed site effects.

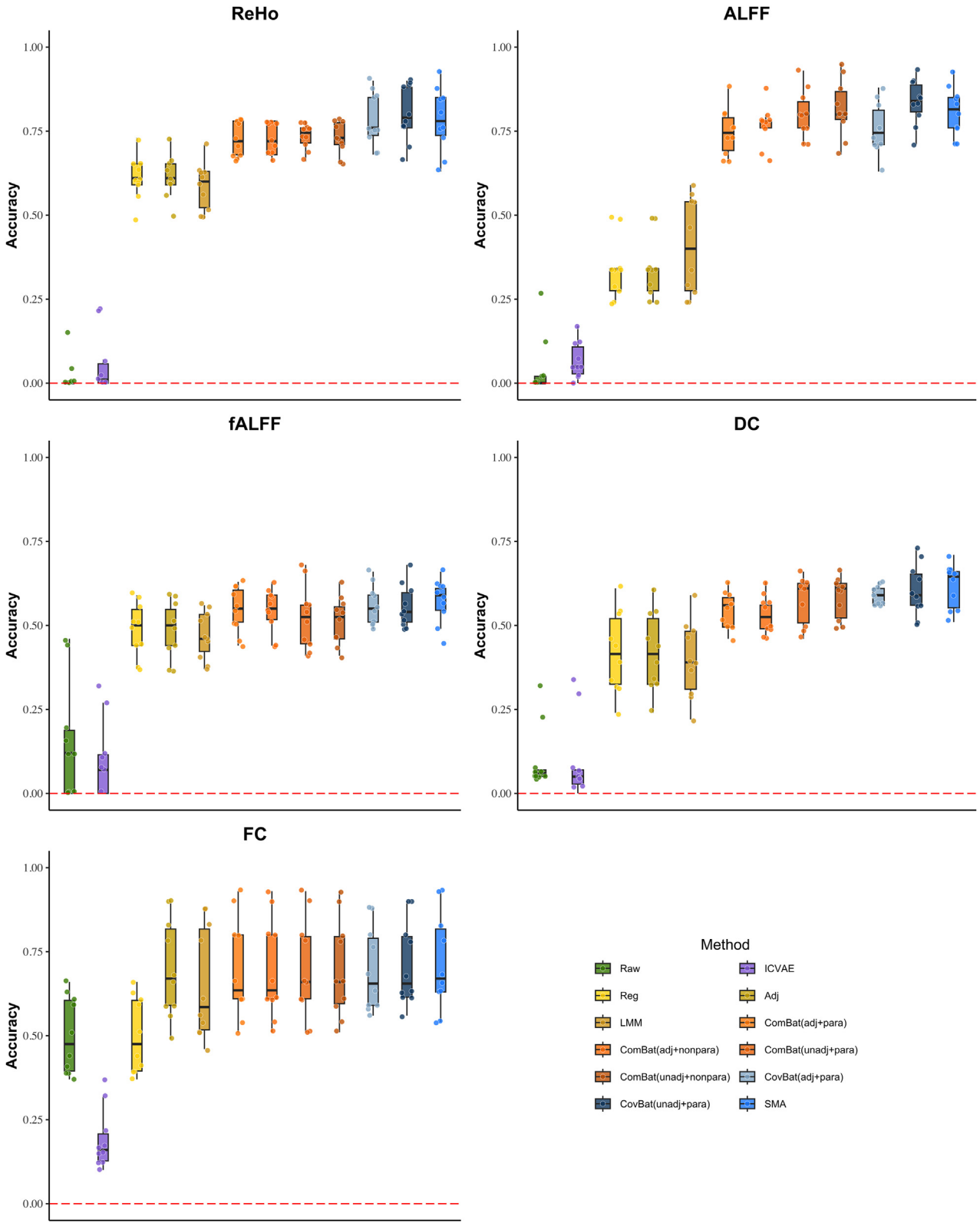


Fig. 4. Identifiability - hierarchical clustering accuracy of TSP-3. For each R-fMRI metric, we applied hierarchical clustering to cluster the traveling subjects' images across three sites to investigate which method could enhance accuracy compared with raw. Out of 41 subjects, the proportion of individuals whose three images from different sites were grouped together is expressed as a ratio. The random chance of accuracy is $\frac{\frac{41}{41}\times 1+\frac{40}{41}\times(S^{1}_{123-40\times 3}-1)+\frac{39}{41}\times(S^{2}_{123-39\times 3}-S^{1}_{123-40\times 3})+\cdots}{S^{41}_{123}}\approx 0$, marked by the red dashed line (S denotes Stirling number, codes to calculate chance level can be found in GitHub). Distances were constructed with 2 algorithms (Euclidean and Pearson) and clustering was based on 5 linkage construction methods (ward.D2, ward.D, average, complete, single). The x-axis indicates the methods and the y-axis presents clustering accuracy; colors distinguish methods and dots refer to 10 values (2 distance algorithms x 5 linkage construction methods). All methods achieved better accuracy than raw except for ICVAE, which failed in ALFF, fALFF, DC and FC (one-tailed, $P < 0.01$). In terms of identifiability, SMA and unadjusted CovBat outperformed the other methods.

Table 4
Dice coefficients for sex differences, for intrinsic brain metrics between two sessions of CoRR after GRF with head motion (mean FD Jenkinson) and age as covariates. The numbers below the percentages represent the number of voxels that were significant in both sessions.

GRF corrected Dice	ReHo	ALFF	fALFF	DC
Raw	0.52[477]	0.61[719]	0.63[3213]	0.50[273]
Reg	0.47[443]	0.65[814]	0.51[1542]	0.42[281]
Adj	0.53[706]	0.66[1128]	0.55[2031]	0.40[399]
LMM	0.51[589]	0.67[985]	0.55[2137]	0.43[344]
ComBat(adj+para)	0.53[734]	0.68[1156]	0.58[2448]	0.41[411]
ComBat(adj+nonpara)	0.53[749]	0.69[1158]	0.59[2506]	0.42[434]
ComBat(unadj+para)	0.48[476]	0.68[976]	0.57[2403]	0.44[301]
ComBat(unadj+nonpara)	0.47[464]	0.69[984]	0.58[2495]	0.43[284]
CovBat(adj+para)	0.55[882]	0.70[1323]	0.58[2559]	0.40[476]
CovBat(unadj+para)	0.48[576]	0.67[1060]	0.55[2132]	0.41[293]
SMA	0.53[802]	0.69[1344]	0.57[2518]	0.41[521]
ICVAE	0.45[736]	0.59[1137]	0.63[5792]	0.51[697]

Table 5
Dice coefficients for CORR session 1, 2, and FCP1/2 with 587 subjects without site as a regressor. The numbers in the brackets represent the numbers of voxels that were significant across datasets.

GRF corrected	ReHo		ALFF		fALFF		DC	
	DiceCoRR1-CoRR2-FCP1	DiceCoRR1-CoRR2-FCP2	Dice CoRR1-CoRR2-FCP1	DiceCoRR1-CoRR2-FCP2	DiceCoRR1-CoRR2-FCP1	DiceCoRR1-CoRR2-FCP2	DiceCoRR1-CoRR2-FCP1	DiceCoRR1-CoRR2-FCP2
Empirical Bayesian ComBat								
ComBat (adj+para)	0.08 [63]	0.07 [59]	0.14 [148]	0.13 [127]	0.13 [234]	0.16 [266]	0.09 [43]	0.09 [40]
ComBat (adj+nonpara)	0.08 [65]	0.07 [62]	0.13 [135]	0.12 [119]	0.12 [222]	0.02 [240]	0.10 [45]	0.08 [35]
CovBat (adj+para)	0.08 [80]	0.07 [69]	0.16 [190]	0.16 [188]	0.04 [60]	0.04 [67]	0.11 [52]	0.00 [0]
Maximum mean distance								
SMA	0.12 [134]	0.11 [130]	0.12 [211]	0.12 [209]	0.19 [466]	0.20 [512]	0.10 [78]	0.09 [70]

FDR corrected	ReHo		ALFF		fALFF		DC		FC	
	DiceCoRR1-CoRR2-FCP1	DiceCoRR1-CoRR2-FCP2	DiceCoRR1-CoRR2-FCP1	DiceCoRR1-CoRR2-FCP2	DiceCoRR1-CoRR2-FCP1	DiceCoRR1-CoRR2-FCP2	DiceCoRR1-CoRR2-FCP1	DiceCoRR1-CoRR2-FCP2	DiceCoRR1-CoRR2-FCP1	DiceCoRR1-CoRR2-FCP2
Empirical Bayesian ComBat										
ComBat (adj+para)	0.16 [320]	0.17 [426]	0.13 [311]	0.13 [285]	0.29 [1649]	0.29 [1654]	0.10 [111]	0.09 [85]	0 [0]	0.01 [2]
ComBat (adj+nonpara)	0.15 [331]	0.15 [310]	0.13 [285]	0.12 [268]	0.30 [1758]	0.30 [1783]	0.10 [120]	0.09 [92]	0 [0]	0.01 [2]
CovBat (adj+para)	0.17 [414]	0.17 [438]	0.15 [386]	0.13 [393]	0.02 [88]	0.06 [250]	0.09 [83]	0.01 [7]	0 [0]	0 [0]
Maximum mean distance										
SMA	0.18 [578]	0.18 [555]	0.19 [801]	0.19 [775]	0.35 [2462]	0.37 [2643]	0.17 [401]	0.17 [401]	0 [0]	0.04 [1]

each subsampling factor between target and source according to Theorem 1 (Zhou et al., 2018)].

We then applied the Friedman Test on replicabilities regarding between-subject sex on five metrics by two multiple correction strategies (GRF and FDR), with post-hoc tests corrected by Tukey's honest significant difference criterion ($P < 0.05$). Raw had a high number of overlapping voxels, but with statistically lower Dice than SMA across metrics. It is true that raw had high number of overlapping voxels and high Dice for fALFF, which we speculate was induced by the large site effects remaining in raw (Fig. 2). SMA was ranked significantly higher than adjusted ComBat/CovBat in both Dice coefficient and the number of overlapped voxels (Fig. 6.B).

3.5. The toy model: Perfect confounding between site and sex

Furthermore, we created an extreme condition. We choose 106 females from the Beijing site and 74 males from the Cambridge site to create a dataset where site and group differences were "perfectly" confounded and used CoRR-SWU (105 female, 116 male) for harmonization. We tested all ComBat series, CovBat, and SMA on ALFF (Fig. 7). We chose replicable clusters as the standards. Since overlapping areas on surface were consistent across different methods (Fig. S6), we take SMA as the reference given it has most overlapping voxels. The outcome of adjusted ComBat and CovBat, which passed GRF correction for both sessions, showed greater similarity to the shared region of two sessions in CoRR-SWU, but did not bear any resemblance to the replicable clusters. SMA results showed overlapped significance in PCC was in line with reproducible results across CoRR and FCP (FCP1/2-CoRR1/2).

In summary, SMA's performance was the best in maintaining individual identification and achieving higher replicability. Unadjusted ComBat/CovBat performed poorly in detecting group-level distinctions, possibly for improper removal of group-confounded site effects. Low replicability Dice values illustrate that PACB, NACB and parametric adjusted CovBat transported discovered patterns across datasets poorly. Moreover, SMA was able to handle extreme conditions in which site and group differences were highly confounded, producing stably reproducible results between sessions and datasets. SMA was better able to harmonize R-fMRI given the above results. Given that applying SMA still entails many degrees of freedom, we executed experiments based on the above datasets.

3.6. Optimization of SMA Target Site Choice

Although SMA outperforms all other strategies, one remaining issue is determining the "target site". Taking advantage of the TSP-3 dataset, we sought to identify how alteration of target site affects identifiability. We directly utilized SMA with distribution shifting but without subsampling subjects given they are the same travelling subjects, with the three sites taking turns as the target site. We evaluated identifiability fluctuations depending on the target site selected; accuracy results are shown in Table S2 (we also demonstrate the results of adjusted ComBat/CovBat

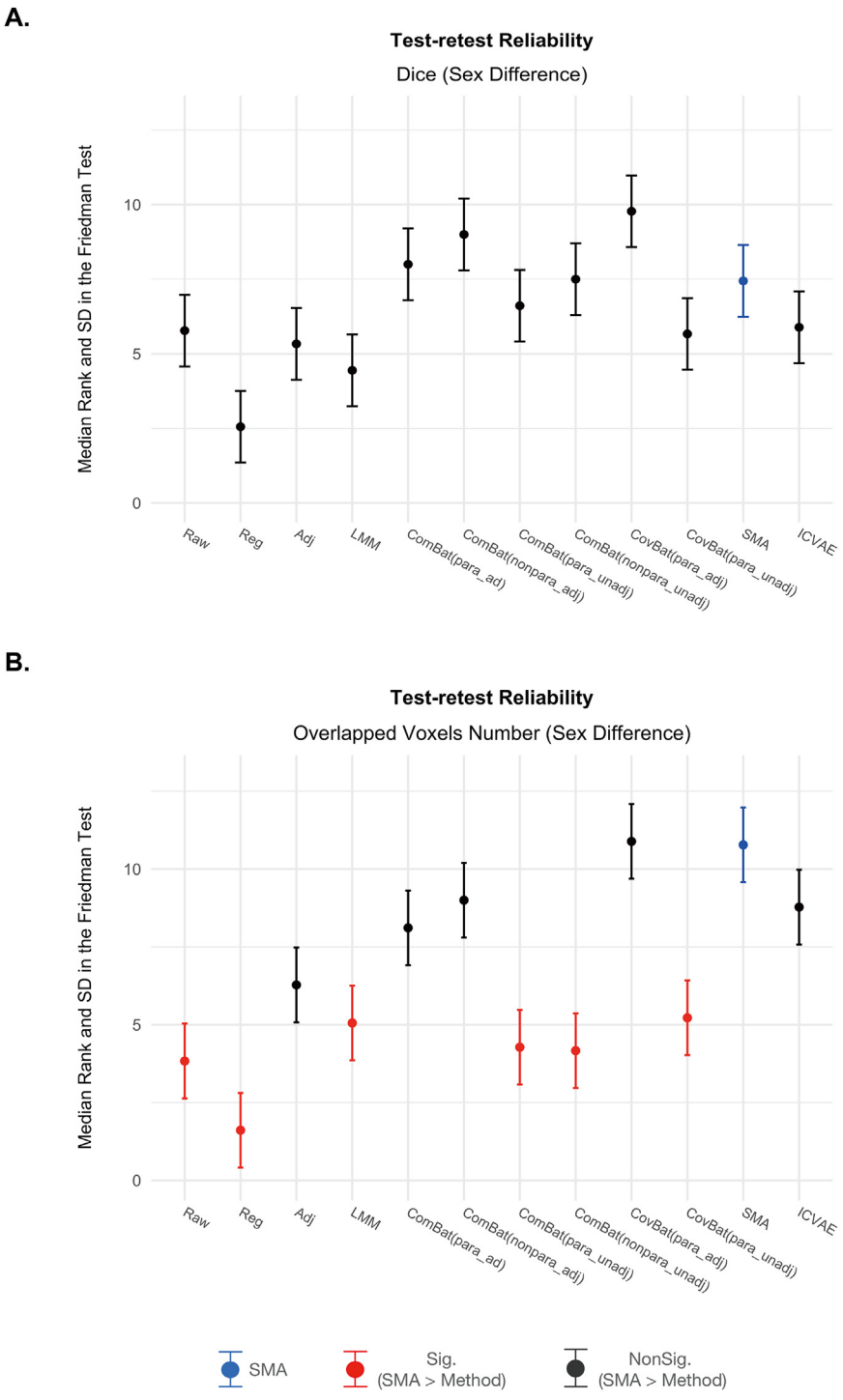


Fig. 5. Statistical comparison on test-retest reliability regarding between-subject sex differences on five metrics by two multiple correction strategies (GRF voxel $P < 0.001$, cluster $P < 0.05$ and FDR $q < 0.05$). We applied the Friedman Test to examine if the differences in test-retest reliability were significant between SMA and other methods from two aspects: (A) Dice coefficients (B) Number of overlapping voxels. The results were graphed with the y-axis showing the rank of each method on test-retest reliability, and the x-axis ordered by methods. SMA is depicted in blue, and methods that differed significantly from SMA in reproducibility are depicted in red (multiple comparison corrected by Tukey's honest significant difference criterion, $P < 0.05$). The high test-retest reliability of the raw data was attributed to site effects. The unadjusted ComBat/CovBat methods were found to be less effective than the adjusted CovBat and SMA methods. The adjusted ComBat/CovBat and SMA methods ranked highly in both aspects of test-retest reliability, indicating their relative effectiveness.

together for comparison). As shown in Fig. 8, in most cases, SMA performed better than or about the same as adjusted ComBat/CovBat regardless of the target site, although fALFF and DC were more sensitive to variations in mechanical properties. To assess statistical significance, we calculated accuracies for IPCAS-GE targeting fALFF and DC and compared them with adjusted ComBat/CovBat ($P < 0.01$, one sided). Results show that SMA was comparable to adjusted ComBat/CovBat for DC, but not as effective as IPCAS-GE for fALFF in identifiability. Although fALFF was limited in identifiability, with a best score of 68% across all methods, users should be aware of this information.

To investigate if target site choice would change test-retest reliability, we varied CoRR target sites and compared the mean Dice coefficients. Specifically, we first employed SMA with different target sites, following the same subgrouping rules. Then we examined sex differences by T-tests on the harmonized results, applied the same correction method and threshold settings, and calculated the Dice coefficients for the two test-retest sessions (test-retest reliability). As displayed in the left part of Fig. 9 and Fig. S5, test-retest reliability fluctuated depending on target site choices and R-fMRI metrics. ReHo, ALFF, and DC changed unanimously across distinct target sites: higher Dice values were found for large sample sites. For fALFF, the regularity isn't that obvious. Overall, larger sample sizes were associated with higher test-retest reliability (BNU 110 subjects, SWU 211 subjects). The dashed lines here refer to the average Dice coefficients of adjusted ComBat/CovBat series, which

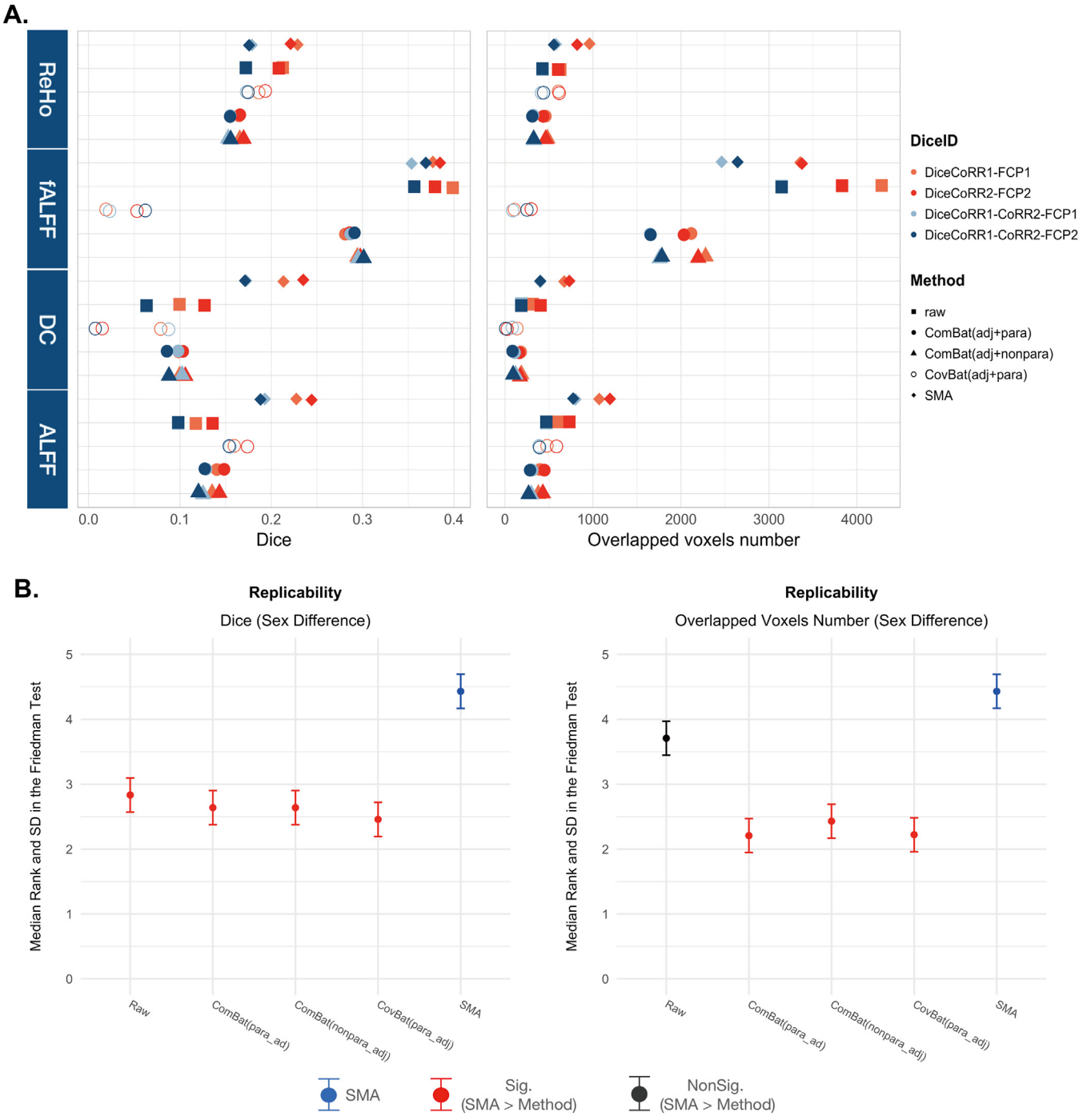


Fig. 6. Replicability. (A) FDR corrected replicability. We calculated the Dice coefficients between CoRR and FCP to check how harmonization affects downstream replicability. The results shown are from methods that performed well both in identifiability (Fig. 4) and test-retest reliability (Fig. 5). Left panel is Dice values and right is the number of overlapping voxels of replicable results across CoRR and FCP, ($q < 0.05$, FDR-corrected). The shape of each point indicates the method used, and the color distinguishes different Dice targets. Raw had a high number of overlapping voxels and high Dice for fALFF, which we speculate was induced by the large site effects remaining in raw (Fig. 2). Among the harmonization methods, SMA showed the highest Dice and number of overlapping voxels in the replicability evaluation. Adjusted CovBat performed well in ReHo and ALFF but failed to achieve similar results in fALFF and DC. (B) Statistical comparison on replicability regarding between-subject sex differences on five metrics by two multiple correction strategies (GRF voxel $P < 0.001$, cluster $P < 0.05$ and FDR $q < 0.05$). We applied the Friedman Test to examine if reproducibility differed significantly between SMA and other methods. SMA is depicted in blue, and methods that differed significantly from SMA in replicability are depicted in red (multiple comparison corrected by Tukey's honest significant difference criterion, $P < 0.05$). Raw had a high number of overlapping voxels, but with statistically lower Dice than SMA across metrics.

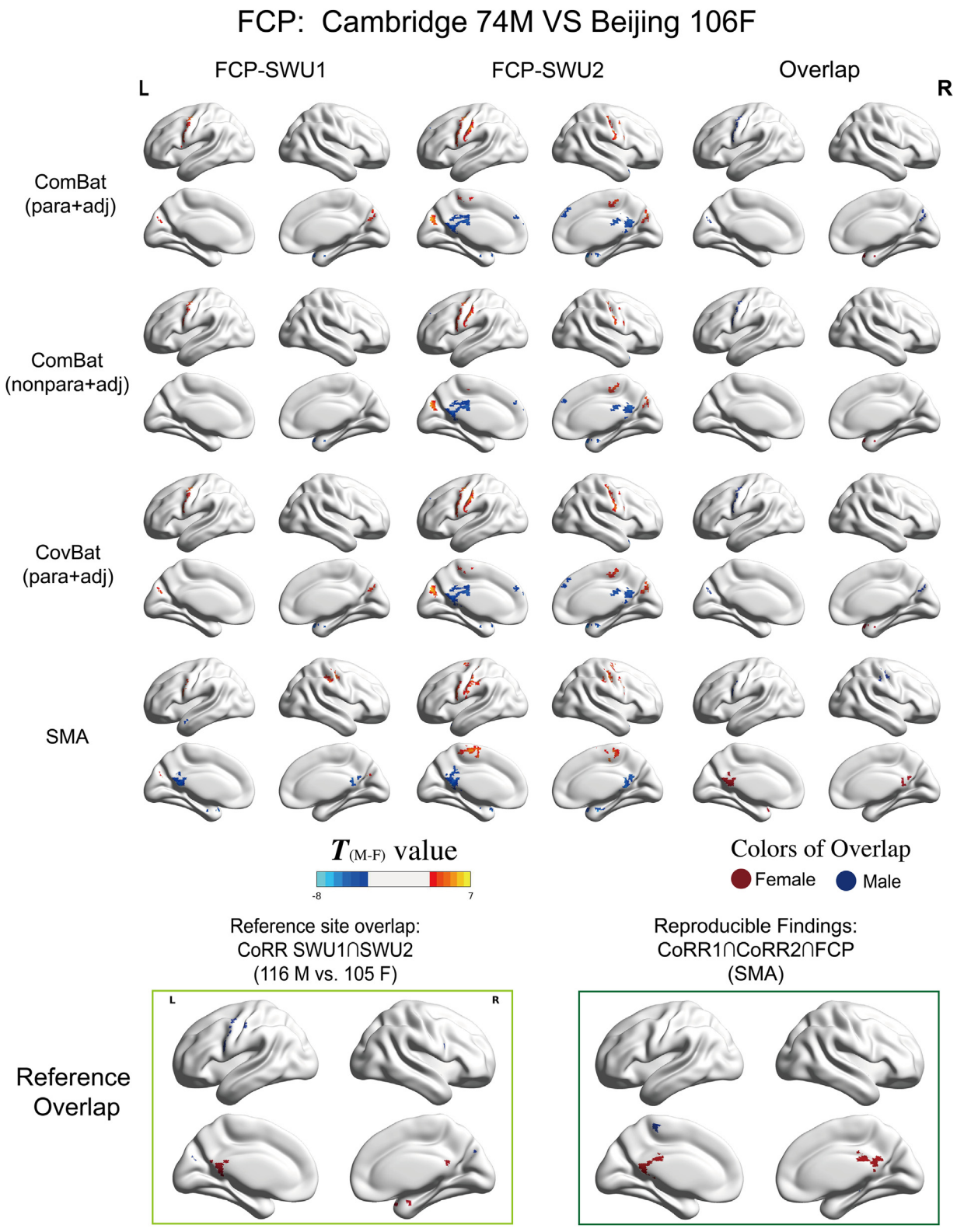


Fig. 7. Evaluation when site and sex confounded. We compared the sex differences of ALFF between 74 males from FCP-Cambridge and 106 females from FCP-Beijing, which were harmonized with ComBat, CovBat, and SMA. For ComBat and CovBat, CoRR-SWU (116 M vs. 105 F) data were harmonized together as a reference. For SMA, CoRR-SWU was used as the target site. Statistical analyses were performed with male-female comparisons and corrected using GRF (voxel $P < 0.001$, cluster $P < 0.05$). The first and second columns display the statistical results for the subjects selected from FCP and harmonized with sessions 1 and 2 of the CoRR SWU site (corresponding scale is at the bottom-middle of the first two columns). Their overlap is shown in the third column. We also provide reference overlaps, on the left the overlap of two sessions in the CoRR SWU site (outlined in grass green) and on the right the reproducible regions across two sessions of CoRR and FCP (outlined in dark green). The replicable regions of other methods are highly similar to SMA but with fewer voxels; details are shown in Table 5 and Supplementary Fig. 6. Red indicates female > male in the overlapping areas and blue is the converse. Only SMA captured the reproducible larger ALFF value of females in PCC in the extreme case.

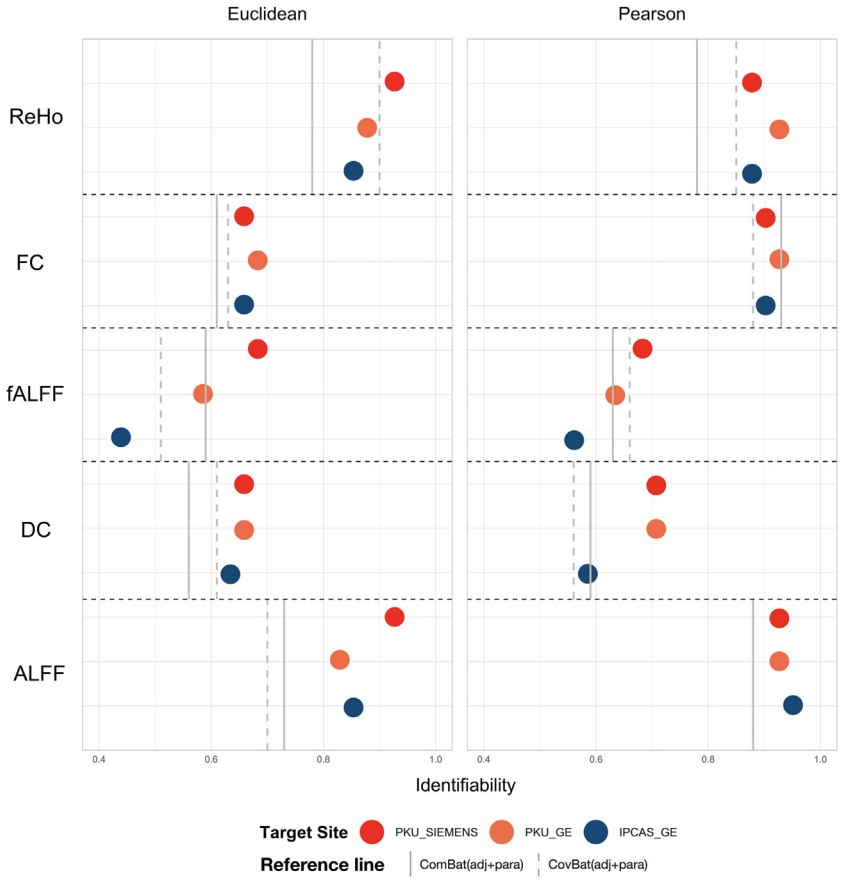


Fig. 8. Site choice experiments – Identifiability on TSP-3. We wanted to see whether changing the target site would affect identifiability significantly. Thus, we computed identifiability accuracy with different sites as the target for TSP-3. The colored points indicate the target site. We label the identifiability of parametric adjusted ComBat and CovBat by solid light grey line and dashed line, respectively. In most cases, SMA performed better than or about the same as adjusted ComBat/CovBat regardless of the target site. However, changing target sites significantly affected the identifiability of fALFF.

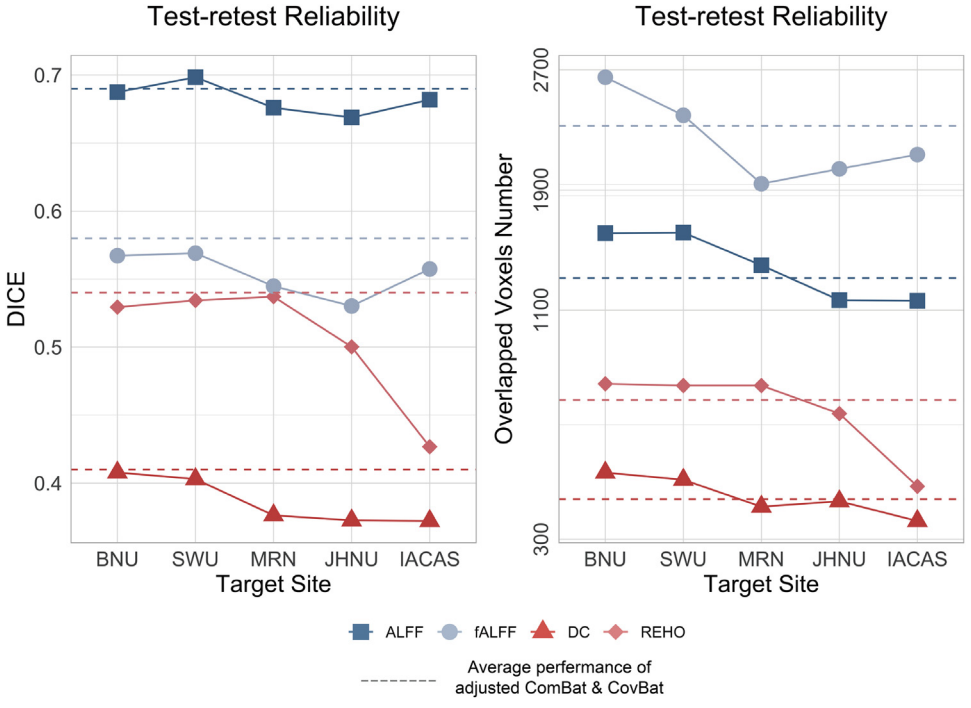


Fig. 9. Site choice experiments – Test-retest reliability on CoRR. We wanted to see how test-retest reliability changed when different sites were chosen as the target. The x-axis is arranged in order by values of the site selection formula, from left to right, from smallest to largest. Color and shape of points distinguish R-fMRI metrics. Dashed lines indicate the average Dice for adjusted ComBat and CovBat. The Dice and number of overlapping voxels of ComBat and CovBat are similar (See Table 4), so averaged their values. Only GRF corrected results are displayed here. In most instances, we found a consistent and noticeable decline in performance from BNU and SWU down to MRN, JHNU, and IACAS. We also noted that the tail of each metric was below the average performance of adjusted ComBat/CovBat. This should be noted by users when facing difficulties in choosing the target site, such as when the sample sizes of all sites are small or overlap of Z variates is limited and there are no additional datasets appropriate to be the target. In such cases, adjusted ComBat/CovBat are good options to consider.

did not have the target site choice issue. We note that they had comparable performance with SMA in test-retest reliability across metrics. This should be noted by users when facing difficulties in choosing the target site, such as when the sample sizes of all sites are small or overlap of Z variates is limited and there are no additional datasets appropriate to be the target. In such cases, adjusted ComBat/CovBat are good options to consider.

We also tested if the results of one target site were more stable than other sites, i.e., if the results were more similar for one targeted site compared to the others using the same data. Using CoRR, we calculated the overlap (Dice coefficient) with results when setting each of the other sites as the target site. Then we averaged the Dice values across the pairs and two sessions for each site [i.e., the stability of site BNU equals the mean of 8 Dice values, e.g., BNU1 ∩ SWU1, BNU1 ∩ MRN1, BNU1 ∩ JHNU1, BNU1 ∩ IACAS1, BNU2 ∩ SWU2, BNU2 ∩ MRN2, BNU2 ∩ JHNU2, BNU2 ∩ IACAS2 (BNU is the target site abbreviation and 1/2 refers to the harmonized CoRR session)]. Supplementary Table 3 shows a descending trend across all R-fMRI metrics, starting from the sites with relatively larger samples (BNU and SWU) and ending at MRN, JHNU and IACAS. While SWU had a sample size twice as large as BNU, the age range was narrower; the age distribution of IACAS was particularly distinct from other sites. Furthermore, we tested whether sample size and Z variate distributions really mattered by constructing different sample size/Z variate distributions of target sites by resampling from the FCP Beijing sample. Specifically, we set three sample sizes (18, 70, 130 subjects) with the same subgroup proportion; we set three Z variate distributions (left bias, origin and right bias) with sample sizes all equal to 70. Each condition was constructed 50 times, with Leiden 2200 and Saint Louis as source sites and harmonized 50 times. Then for each harmonized result, we employed sex difference statistics and computed Dice coefficients for each pair of GRF corrected results (voxel $P < 0.001$, cluster $P < 0.05$). The results are shown in Fig. 10. To summarize, both sample size and Z distribution influenced the stability of harmonized results. Still, when the conditions were not too far away from “the ideal” (like Left bias – origin, 70 – 130), stabilities were close to 1. In the worst cases we created here (Right bias, sample size 18) stabilities had relative larger variations yet still produced relative high average Dice coefficients (around 0.90 and 0.87, respectively).

These phenomena indicate that both sample size and subgroups’ distribution between source sites and the target site modulate harmonization and downstream analysis. The larger the sample size of a target, the better the test-retest reliability. It is also beneficial that the subgrouping factor distribution (Z variates) of a target site can cover that of as many source sites as possible. If a site is a better target, its demography would need the fewest extra samples to reproduce the others (i.e., source sites). Given the above experience we provide a heuristic formula for target site selection:

$$\min\left[\frac{1}{n_j^{subjects}}\sum_{i\neq j,\, i,j\subseteq N} D_{KL}(P(Y_i)||P(Y_j))\right]$$

where Y represents subgroup labels, j for target site ID, and i for source site ID. $P(Y_i)$ and $P(Y_j)$ stand for subgroup percentages in source site and target site. D_{KL} refers to Kullback-Leibler Divergence calculator. Here, it quantifies the amount of extra information for site j needed to characterize the population differences and sample selection of site i, based on the corresponding sample features of site j. N means the set of site IDs and n_j means the number of subjects in site j. Finding the minimum value is equivalent to finding which site needs the least extra information to encode all other sites in terms of the relevant demographic factors, while prioritizing relatively large sample size. When utilizing this formula, we get the best target site is BNU, followed by SWU, MRN, JHNU and IACAS in descending order. Here we do not consider data quality affected by hardwares etc., which should be taken into consideration given to the specific situation.

Based on the above tests, we provide the following recommendations:

1 **Site chosen**: The ideal situation is a large sample-size site with relatively even distributions, which means sample numbers in each division are comparable to those of source sites, thus reducing bias induced by sample selection. If such a site does not exist, we prefer the largest sample-size site containing enough demographic range (such as BNU or SWU) in both continuous and discrete biological factors. Suboptimal choices are those with moderate sample size but even demographic distribution (such as MRN). We provide an empirical target selection formula for reference.
2 **Subsampling**: This mainly depends on the hypothesized relationships among factors. Zhou et al., (Zhou et al., 2018) used a graphical causal model to depict the causal relationships among age, diagnosis and other endogenous variables influencing CSF measurements. In our example, we regard both sex and age as impacting R-fMRI metrics and demographic characteristic difference E_P and selection bias E_B directly affecting age and sex. Here we take 21 years old as the age division threshold based on existing evidence.

Briefly, a site with large sample size, whose Z variates distribution is either close to the study-target population or the closest to the rest of the pooling sites would be an ideal target site. Our proposed formula can be easily used to determine the suitable target site.

4. Discussion

To evaluate appropriate brain imaging harmonization methods for the big data era, we first checked whether proposed techniques can remove bias of mean, variance and covariance of site effects. ICVAE reduced but did not eliminate significant site effects over cortex for TSP-3 and CoRR; SMA retained some significant subcortical site effects in the FCP data. Other methods performed well on eliminating mean and variance bias. Still SMA bested the others in hiding site labels from a ML algorithm. Next, we simplified the challenge by examining the same participants traveling amongst three scanners. When evaluating identifiability, SMA and parametric-unadjusted CovBat were the top performers with superb control of bias and variances. In the next step, we examined test-retest reliability of biological differences in multisite large sample datasets with each participant scanned twice on one scanner. SMA and parametric-adjusted CovBat produced the largest number of overlapping voxels. Next, generalizability across datasets was enhanced by SMA, excelling others in Dice coefficient and number of overlapped voxels. For example, SMA-harmonized detected higher PCC activity in females for both ALFF and fALFF. By contrast, ComBat-harmonized could not transport findings from CoRR to FCP. Finally, SMA stably reproduced ALFF sex differences in the sex-site confounded scenario, whereas results of adjusted ComBat/CovBat harmonized varied widely when using CoRR-SWU1/2 for harmonization. By leveraging TSP-3, CoRR and FCP, we sought how to best select a target site to optimize harmonization results. Assessments of identification and test-retest reliability show that the site with the largest sample size and widest overlap distribution with all other sites should be selected as the target site. Further we tested and proved that these two factors affect stability significantly with the FCP dataset.

We noted that in the identifiability test, ComBat series’ performance varied across different R-fMRI metrics and detailed harmonization strategies (e.g., parametric/adjusted). Unadjusted ComBat/CovBat series have comparable or even better performance in some metrics on identifiability compared with SMA, but the number of significant overlapped voxels are much fewer in test-retest reliability (Table 4). In replicability, though ComBat/CovBat may seem close with SMA in Dice value, SMA performed significantly better in both the Dice coefficient and the number of overlapped voxels (Fig. 6). If we want to use ComBat/CovBat to harmonize R-fMRI metrics, we need to change our

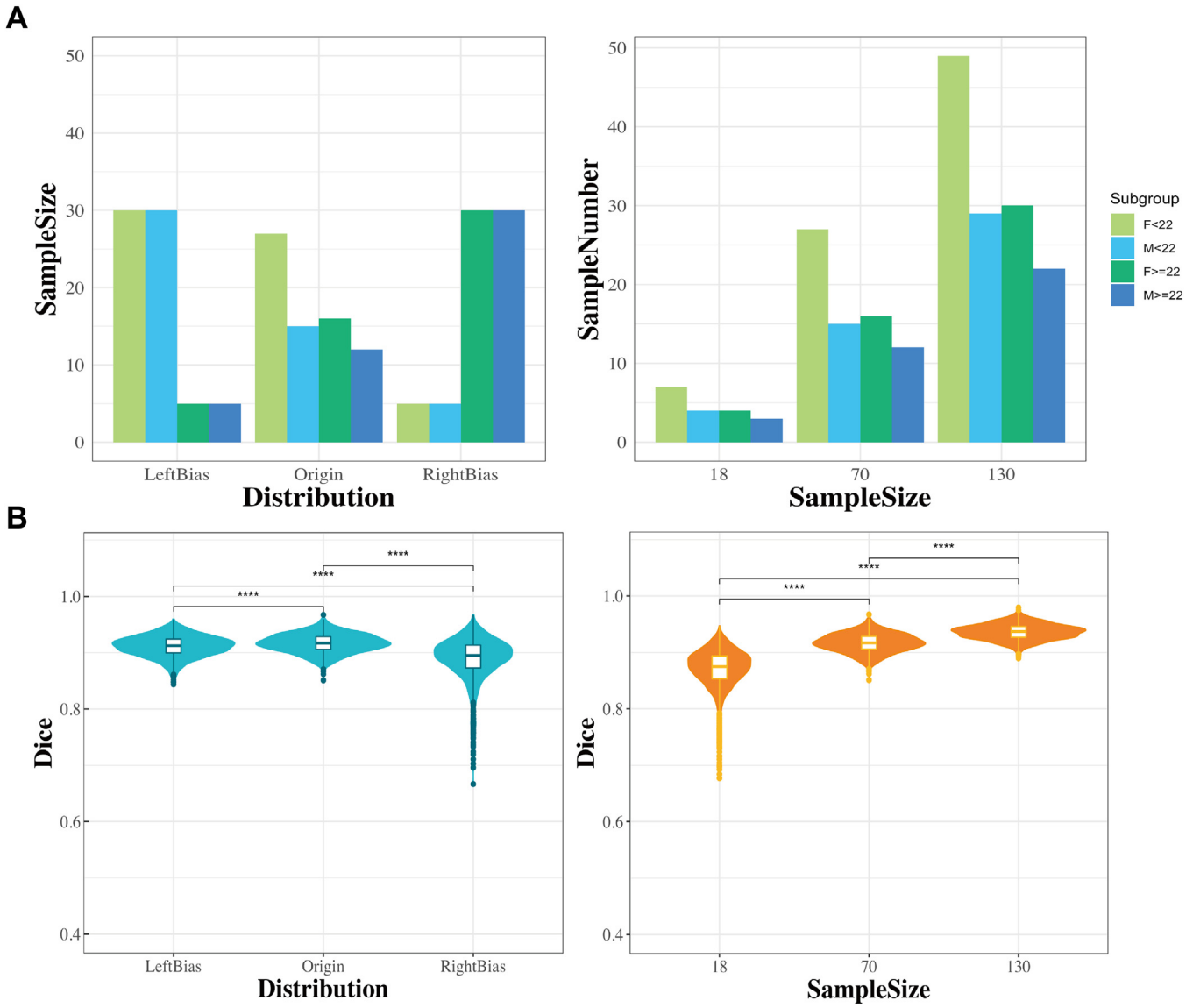


Fig. 10. Target Site choice experiments - Mis-use cases. We tested whether Z variate distributions and sample sizes of the target site affect the stability of harmonized results. (A) The histograms along the x-axis of each subfigure are the distribution/sample size of the target site, with the other factor staying the same across different distributions (70 subjects) /sample sizes (subgroup proportion). (B) The x-axis corresponds to the distribution/sample sizes of the target sites presented in A. The y-axis shows the Dice values of each pair of harmonized statistical maps of sex differences, GRF corrected (voxel $P < 0.001$, cluster $P < 0.05$). The results in B show that an appropriate distribution or a large enough sample size in a target site produces stable harmonization. Still, the influence is negligible between Left Bias and Origin and sample sizes between 70 and 130. We conclude target sites can be safely selected within this range. The target site recommendation formula helps avoid less ideal conditions (e.g. Right bias or samples of 18).

strategies according to different application scenarios, a methodological limitation that does not affect SMA. The ComBat algorithm can be decomposed into a linear part with the same logic as ordinary linear regression and the standard residual variance part, which normalizes each site's error variance by dividing it by the Bayesian-estimated site-wise standard error. The ComBat model can only include linear relationships between independent and dependent factors. In our case, the linear hypothesis impaired identifiability yet enhanced test-retest reliability since dependence between estimates of slope and class value could lead to an artificially increased class signal. Moreover, the way ComBat standardizes sites with a once-for-all estimation of site distribution hyperparameters would diminish the contribution of small-scale sites. The more scanners are involved, especially when most sites have fewer than 30 subjects, the larger the estimated standard deviation of site variance. Consequently, biological diversity among groups could be diminished, resulting in insensitivity to moderate contrasts, reflected in 1) fewer significant voxels compared with SMA with the same number of subjects and 2) extensive test-retest reliability but lower replicability with some significant areas missed.

SMA balances individual-level differences, exemplified by high accuracy in identification clustering, and group-level characteristics, illustrated by stably higher Dice coefficients in both test-retest reliability and replicability. Significantly higher fALFF in males was found in bilateral insula, sensory and motor related cortices, consistent with prior studies (Al Zoubi et al., 2020; Allen et al., 2011). In addition, areas with significant fALFF were highly spatially similar to those reported by Biswal et al. (Biswal et al., 2010), showing high replicability and statistical power after SMA harmonization. What's more, SMA retains ALFF replicable results even when sex and site are totally confounded, no matter which session of CoRR-SWU was used as the target site. The SMA-harmonized approach was the least accurate in site prediction and showed high accuracy in individual identification. These quantified qualities confirm

SMA's preeminent role in R-fMRI site effect harmonization. Accordingly, we elaborate on SMA's advantages in harmonization. On the whole, SMA harmonizes site effects through one-on-one "coaching" (the target site "teaches" the source sites), which prevents biasing toward large-sample sites. Meanwhile, its subsampling algorithm retains enough information of interest and offsets the influence of outliers and sample size. Zooming in on the technical details, the subsampling scheme has two advantages. One relates to "landmark" data (Gong et al., 2013). The landmarks in the harmonization condition are those matched subgroups coordinated by Z variates, corresponding to d-separation variables in a graphical causal model. This trick controls the match of biological and other covariates between sites and keeps features from the source site adaptive to the target site. The other advantage is resampling, which can make the results robust to outliers even with small sample sizes, enhancing generalizability.

Fundamental psychiatric issues such as comorbidity (McGrath et al., 2020) and unpredictable progression of mental disorders (Quinlan et al., 2020) hinder timely diagnosis and precision treatment. The challenge entails relatively low signal intensity specific to group-level psychological/psychiatric disorders in contrast to between-subject variances. Progressing without consistent and valid data as the foundation is not easy. Reaching that goal requires confronting the heterogeneity inherent in synthesizing small-scale data from diverse sources (Yan et al., 2019). The bias and variances among different scanners, selection standards and samples result in inconsistencies among sites, which are nonlinearly expanded along with the number of sources. Solutions responding to this need are emerging in the intersection of medical imaging and artificial intelligence (Dinsdale et al., 2021; Guan et al., 2021; Zhong et al., 2020). However, only a few of these methods were designed for or have experimented with R-fMRI data. Moreover, few methods have thoroughly tackled residual site effects, identifiability, test-retest reliability and replicability, probably due to a lack of appropriate datasets designed for such purposes. Our work fills this gap and supports the conclusion that a nonparametric algorithm fits the above requirements. SMA was inspired by causal inference. A flexible and free application structure empowers it, and can be easily extended and modified to deal with many situations. Still, freedom has a price. The price of this flexibility is precisely what data science requires. We must dig into our data to learn its distribution, dimensions, relationships among factors, and population from where it was sampled as the priors that will guide subsequent analyses.

Some methods had comparable performance with SMA in some aspects, e.g., unadjusted CovBat exceeded SMA in identifiability in ReHo, yet it was unable to reach same level in test-retest reliability or in retaining biological differences when site and sex were fully confounded. Accordingly, we conclude that SMA performed the best overall. Additionally, we offer a heuristic formula that provides a rough recommendation for selecting a target site. Still, these recommendations are based on the limited datasets we assessed in this work. We have opened our data and codes, and an easy-to-use module specialized for harmonization will be released in the next version of DPABI, including SMA, CovBat, ComBat and ICVAE, etc. Together with replication and validation efforts from other researchers, we hope this work will encourage better harmonization methods and allow the field to reach consensus.

Additionally, it is important to note that as the number of sites increases, the use of SMA becomes more convenient. This is because SMA can easily adapt to new sites being pooled only asking for the data from target site. On the other hand, methods like Bayesian-based would require all unharmonized data, which can be problematic for large-scale datasets in terms of storage and processing requirements. Specifying a target site is considered a disadvantage of SMA, but it can also be an advantage in the era of big data. For instance, if a model is trained with a subject pool of 100,000, newly collected data from an unseen site must be harmonized with all 100,000 subjects using ComBat or CovBat. However, with SMA, one only needs to establish a good target dataset of, say, 5,000 subjects from a single site with good age and sex distribution. Then, the newly collected data from unseen sites can be harmonized with only this target dataset using SMA, leading to increased efficiency in generalizing models trained on big data.

Based on our experiments, the replicability of sex was low after multiple comparison correction. A recent paper (Rauch and Eliot, 2022) reported that non-binary gender is as well or better associated with human behavior and brain imaging than binary sex. Those authors noted that "…a recent synthesis of brain imaging and postmortem studies found surprisingly small and poorly reproduced sex differences in structural volumes, laterality, connectivity and fMRI activation, after global brain or head size has been controlled for (Eliot et al., 2021)…". This reminds us that when trying to find reproducible relationships between phenotypes and brain characteristics, both sides need to be defined carefully and precisely. This phenomenon may also explain why traditional psychiatry categories may not accurately partition varying symptoms and their combinations in subjects. When we take psychiatric categories as independent variables, many covariates/confounders can be overlooked. Applying RDoC (Sanislow et al., 2019), as a framework to integrate multilevel information may be a promising path forward.

We note some study limitations. Our analyses were primarily conducted in volume space, while studies have shown surface-based approaches enable more precise spatial localization (Coalson et al., 2018) and decrease signal contamination (Brodoehl et al., 2020). Our findings should be further validated with surface-based preprocessing methods. Moreover, guidelines on choosing the target site for optimizing SMA are preliminary. Our formula provides a preliminary mechanism for filtering target sites which should be further tested and modified to enhance its generalizability.

In the future, it would be preferable to establish a thorough and instructive processing standard for implementing SMA. We welcome and advocate the design and collection of large-scale test-retest datasets consisting of traveling subjects (Yamashita et al., 2019) that would include more recorded variables for verification and increasing improvements. State-of-the-art techniques aimed at harmonizing R-fMRI should be compared to SMA on identifiability, test-retest reliability and generalizability as well as on applicability. For a complicated structured neural network such as ICVAE, a heavy workload is transferred to representative data accumulation and experiments to testify its generalizability. Making codes open and providing clear instruction manuals will advance the translation and application of new AI techniques regardless of users' professional backgrounds.

5. Conclusion

In this study, we selected site-harmonization approaches along the statistical theory spectrum and systematically compared them in validity, test-retest reliability, and generalizability on multiple datasets. SMA is for now the best choice for site harmonization, performing well overall. We offered our application recommendations for SMA based on experiments and suggested future studies to harmonize site effects in brain imaging big data.

Declaration of Competing Interest

The authors declare no competing financial interests.

Credit authorship contribution statement

Yu-Wei Wang: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Funding acquisition. **Xiao Chen:** Investigation, Writing – review & editing. **Chao-Gan Yan:** Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Data availability

All data were obtained from publicly available online resources: TSP-3: (http://rfmri.org/RuminationfMRIData or http://doi.org/10.57760/sciencedb.o00115.00002. CoRR: http://fcon_1000.projects.nitrc.org/indi/CoRR/html/index.html. FCP: http://fcon_1000.projects.nitrc.org/index.html. All codes used in the current study are available online: https://github.com/Chaogan-Yan/PaperScripts/tree/master/WangYW_2023_NeuroImage. All the harmonization methods used herein will be integrated into easy-to-use modules in the next release of DPABI software (http://rfmri.org/dpabi).

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.neuroimage.2023.120089.

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